

CS-736

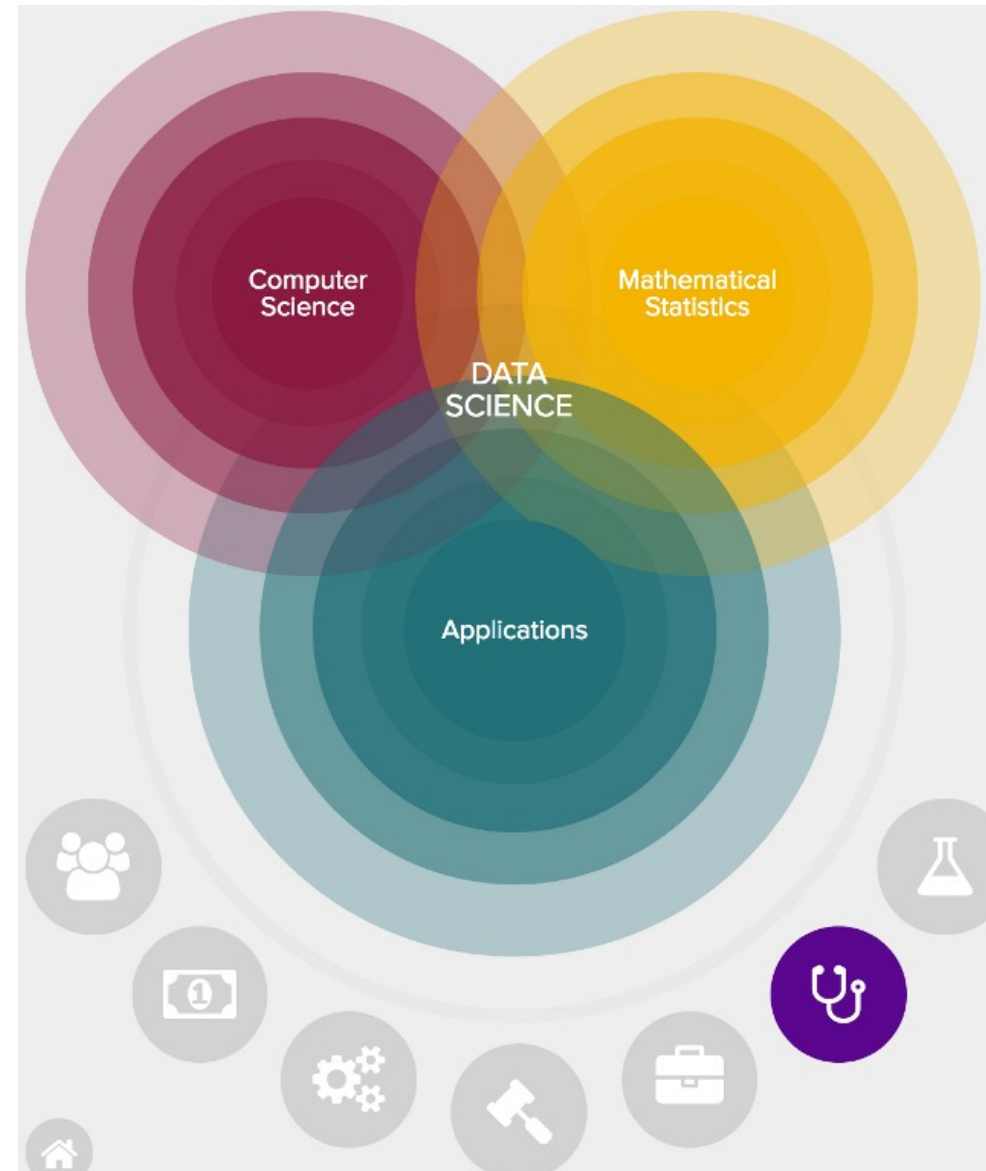
Medical Image Computing (MIC)

Introduction

Suyash P. Awate

Data Science

- As per Wikipedia:
 - “Concept to unify statistics, data analysis, machine learning & their related methods” in order to “understand & analyze actual phenomena” with data
- Spans multiple disciplines
 - Mathematics
 - Statistics
 - Optimization
 - Computing
 - High performance
 - Big data
 - Application domain insights



DATA SCIENTIST

≡≡≡ MUST-HAVE SKILLS ≡≡≡

MATH & STATISTICS

- Machine Learning
- Statistical Modeling
- Exploratory Analysis
- Clustering
- Regression Analysis

DOMAIN KNOWLEDGE & SOFT SKILLS

- Inclination towards business operations
- Keen on working with data
- Problem solver
- Strategic, proactive, and cooperative
- Interested in hacking



PROGRAMMING & DATABASE

- Computer Science Fundamentals
- Database Management System
- Data Visualization
- Python
- Big Data

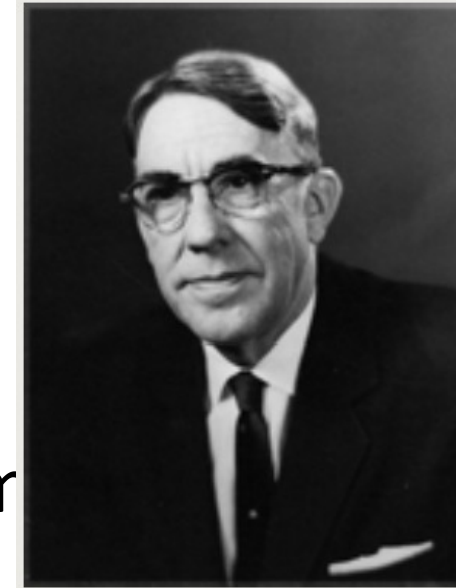
COMMUNICATION & VISUALIZATION

- Storytelling skills
- Convert data-based insights into decisions
- Collaborative with Sr. Management
- Knowledge of tools like Tableau
- Visual art design

Data Science ↔ Machine Learning

- A Samuel [1959] coined “**machine learning**”
 - Field of study that gives computers the ability to learn without being explicitly programmed
 - “Unlike a rule-based system” where experts hard-code domain knowledge directly into system
- M Jordan [2014]
 - Suggested “**data science**” as a placeholder for the broad field
 - Machine learning & statistics are closely related fields.
The ideas of machine learning, from methodological principles to theoretical tools, have had a **long pre-history in statistics.**

Professor Arthur Samuel



Berkeley | EECS
Electrical Engineering and Computer Sciences



Michael Jordan
Professor

Data Science

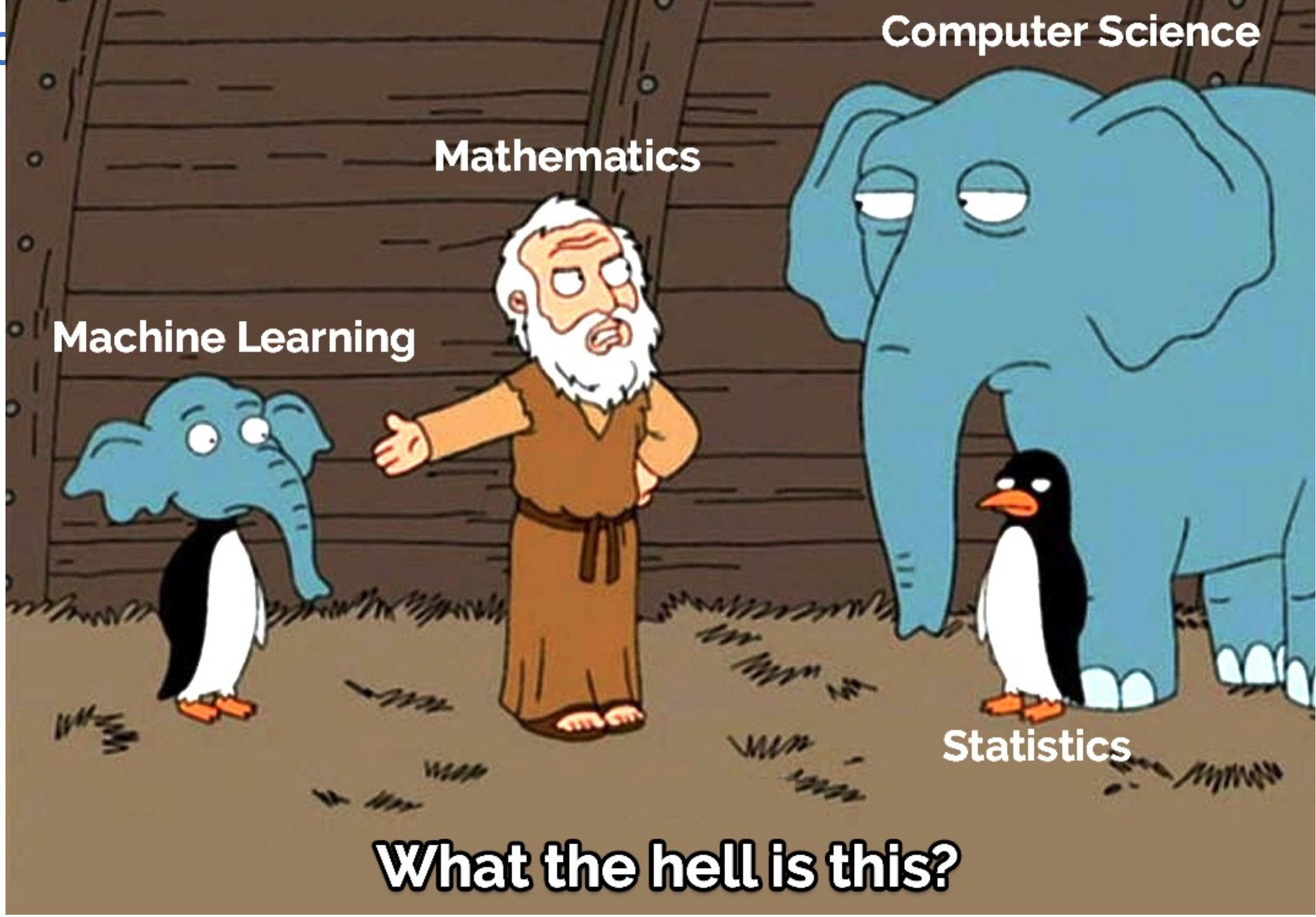
Computer Science

Mathematics

Machine Learning

Statistics

What the hell is this?



The New York Times

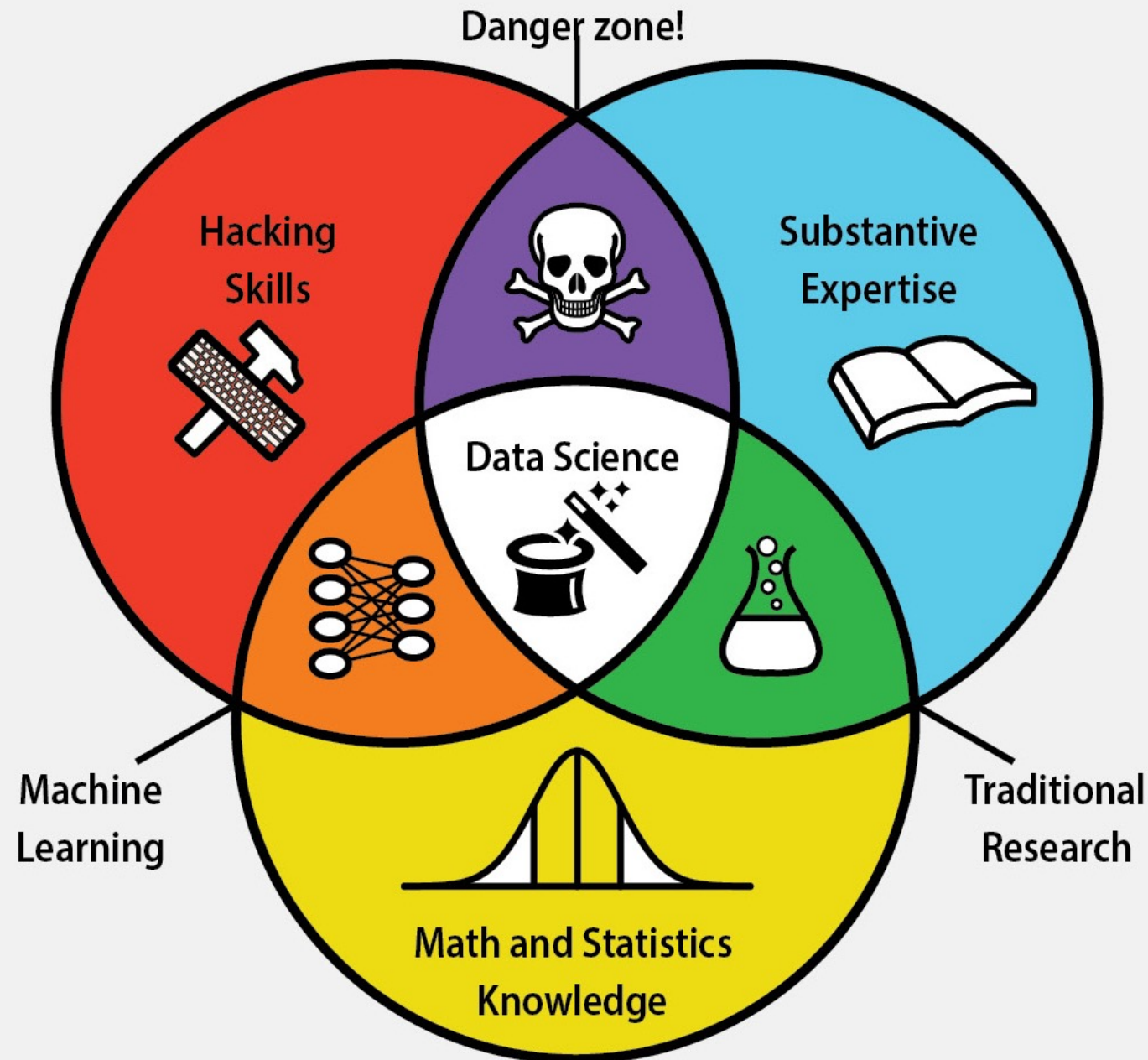
TECHNOLOGY

For Today's Graduate, Just One Word: Statistics

By STEVE LOHR AUG. 5, 2009

“The key is to let computers do what they are good at, which is trawling these massive data sets for something that is mathematically odd,” said Daniel Gruhl, an I.B.M. researcher whose recent work includes mining medical data to improve treatment. “And that makes it easier for humans to do what they are good at — explain those anomalies.”

DATA SCIENCE SKILLSET



Data science, due to its interdisciplinary nature, requires an intersection of abilities: **hacking skills**, **math and statistics knowledge**, and **substantive expertise** in a field of science.



Hacking skills are necessary for working with massive amounts of electronic data that must be acquired, cleaned, and manipulated.



Math and statistics knowledge allows a data scientist to choose appropriate methods and tools in order to extract insight from data.



Substantive expertise in a scientific field is crucial for generating motivating questions and hypotheses and interpreting results.



Traditional research lies at the intersection of knowledge of math and statistics with substantive expertise in a scientific field.



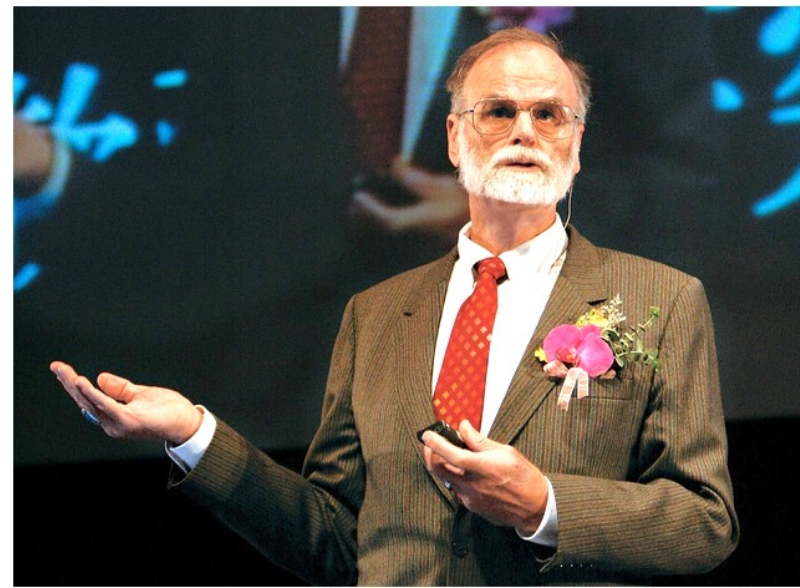
Machine learning stems from combining hacking skills with math and statistics knowledge, but does not require scientific motivation.



Danger zone! Hacking skills combined with substantive scientific expertise without rigorous methods can beget incorrect analyses.

Data Science ↔ Data

- Jim Gray:
Turing Award winner
- Paradigms of science
 - 1) Experimental science
 - 2) Theoretical science
 - 3) Computational science
 - 4) **Data-driven / data-intensive** scientific discovery
 - Data exploration and hypothesis building itself relies on computing on massive data



The FOURTH PARADIGM

DATA-INTENSIVE SCIENTIFIC DISCOVERY

EDITED BY TONY HEY, STEWART TANSLEY, AND KRISTIN TOLLE

**Harvard
Business
Review**

Data Scientist: The Sexiest Job of the 21st Century

by [Thomas H. Davenport](#) and [D.J. Patil](#)

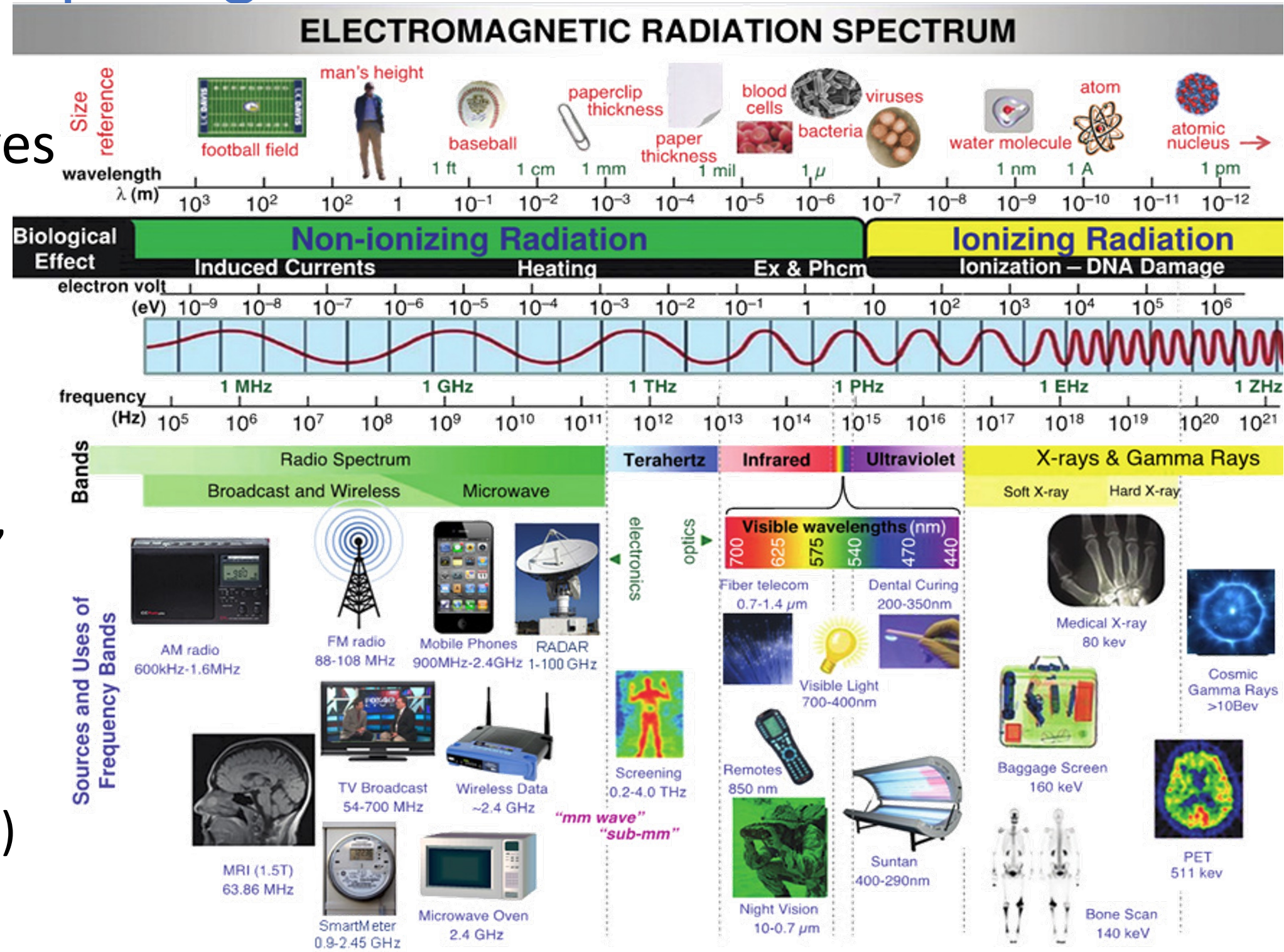
From the October 2012 Issue

Medical Image Computing

- This field develops computational and mathematical methods for solving problems pertaining to medical images and their use for biomedical research and clinical care [Wikipedia]
- Medical image computing is an interdisciplinary field at the intersection of:
 - 1) Computer science, information engineering, electrical engineering
 - Algorithms, computing, hardware engineering
 - 2) Mathematics and Physics
 - Physics: Model processes of scanner measurement / sensing / imaging
 - Mathematics: Model data, learn models from data
 - 3) Medicine
 - Applications, driving problems in science and clinic

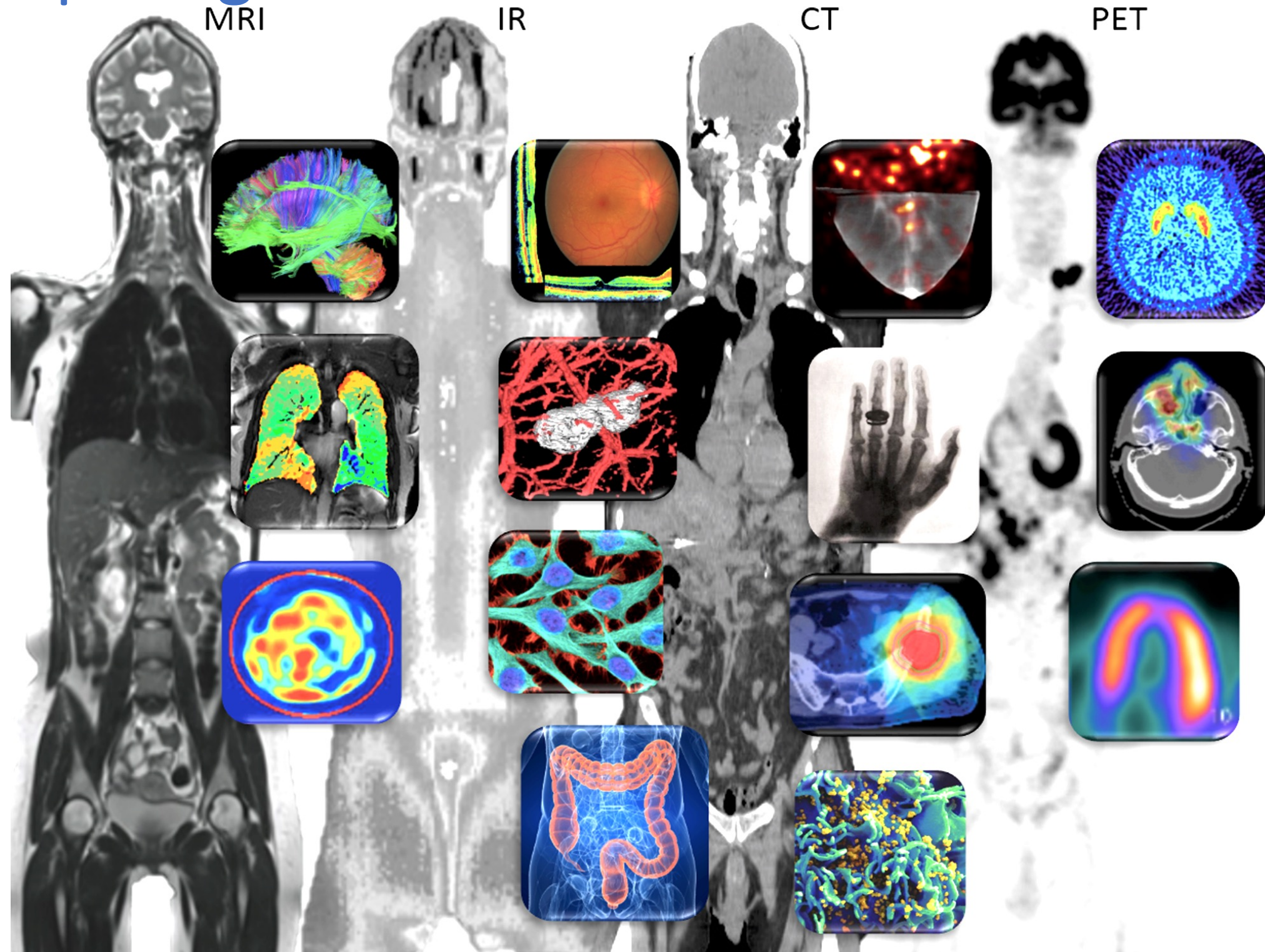
Medical Image Computing

- Imaging
- Electromagnetic waves
 - Ionizing radiation: X-rays (CT), gamma rays (PET, radiation therapy)
 - Visible spectrum: camera (endoscopy), light microscopy
 - Infra-red waves (thermal imaging)
 - Radio spectrum: magnetic fields (MRI)
- Sound waves



Medical Image Computing

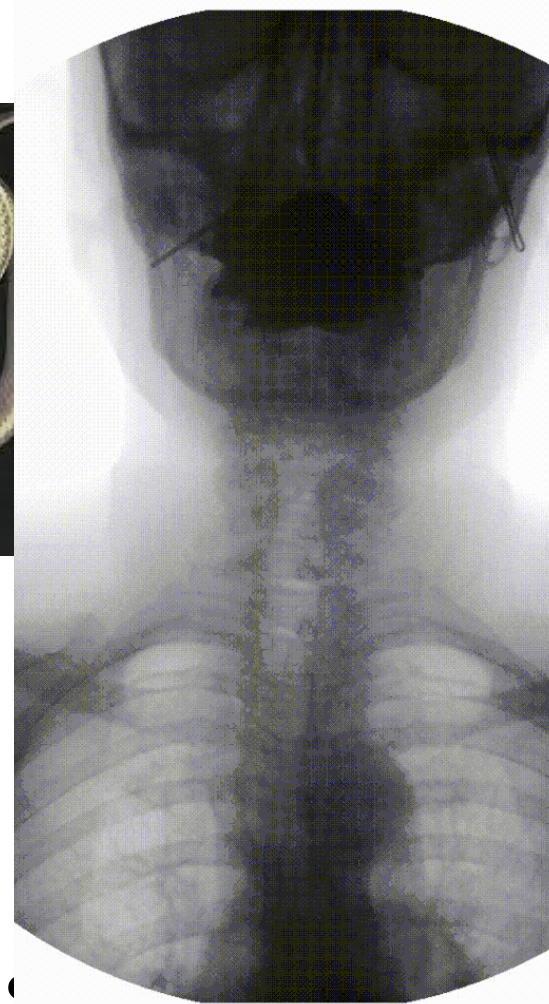
- Imaging
- Modalities
- Tissue properties
 - Photographs
 - Density
 - Magnetism
 - Chemical content
 - Diffusivity
 - Metabolism
 - Temperature
 - Cells
 - ...



Medical Image Computing

- **History**

- X-ray: 1895
 - Film invented by Kodak 1918
- Contrast agent + X-ray fluoroscopy: 1910
 - Seeing object motion
- Ultrasound: 1940s
- Positron emission tomography (PET): 1950s
- Nuclear magnetic resonance (NMR): 1950s
- Magnetic resonance imaging (MRI): late 1970s
- X-ray computed tomography (CT): early 1970s
- Radiation therapy for cancer (gamma rays, ...)



Medical Image Computing (MIC) CS-736

- CS-736 focuses on aspects related to computer science & engineering
 - Image analysis
 - Machine learning: statistical modeling and inference
 - Algorithms
 - Image data + Mathematics + Computing
- Data = digital image (multi-dimensional array of numbers)
- Medical Imaging
 - Modalities: Xray, CT, MRI, ultrasound, microscopy, ...
 - Applications are ubiquitous: bones, soft organs (lungs, brain, ...), cells, ...
- Algorithms
 - Input = image
 - Output = image / measurement / high-level description

Medical Image Computing (MIC) CS-736

- Comparison with other fields in data analysis
 - Some overlap in principles
 - Robot vision, remote sensing, astronomy, seismic, underwater, audio, text, ...
- What's different about medical images ?
 - Acquired image data can have a specific structure
 - What physical entity ? What kind / process of measurement ?
 - Image degradation / noise / [artifacts](#) have a specific structure
- What's different about algorithms ?
 - Clinical / scientific applications need specific kinds of information
 - This course focuses on principled algorithm design
 - Mathematical modeling
 - Formulation as an optimization problem
 - Image-processing algorithm = solver for optimization problem

Medical Image Computing (MIC) CS-736

- Why take the course ?

- 1) Research

- Image computing, data science, machine learning, ...

- 2) Learning and application of (evergreen) concepts from:

- Statistics, optimization, machine learning

- 3) Jobs in industrial R&D

- India

- GE, Siemens, Philips
- Samsung, Canon / Toshiba, Sony
- Sun Pharma
- Agilent (HP), Intel, ...
- Startups

- Abroad: Microsoft, Nvidia, Toshiba, Medtronic, Hitachi, Pharma., Google, ...

Medical Image Computing (MIC) CS-736

- Theory
 - Where do existing algorithms come from ?
 - How do we design new algorithms for new real-world tasks ?
 - How do we leverage prior information learned through experience ?
 - Why do certain algorithms work ? Work better on some data than others ?
- Theoretical areas will assume background in:
 - Basics (IITB-CSE UG level) of:
 - Linear algebra, optimization, graph theory
 - More than basics of:
 - Probability, statistics
 - Some basics of image processing:
 - Convolution, filtering, Fourier transforms

Medical Image Computing (MIC) CS-736

© Randy Glasbergen / glasbergen.com

- Interactions
- **moodle.iitb.ac.in**
 - Receiving (and submitting) assignments and other learning materials
 - Receiving grades, feedback, comments
 - Communications



“You have to attend classes. You can’t just follow me on Twitter.”

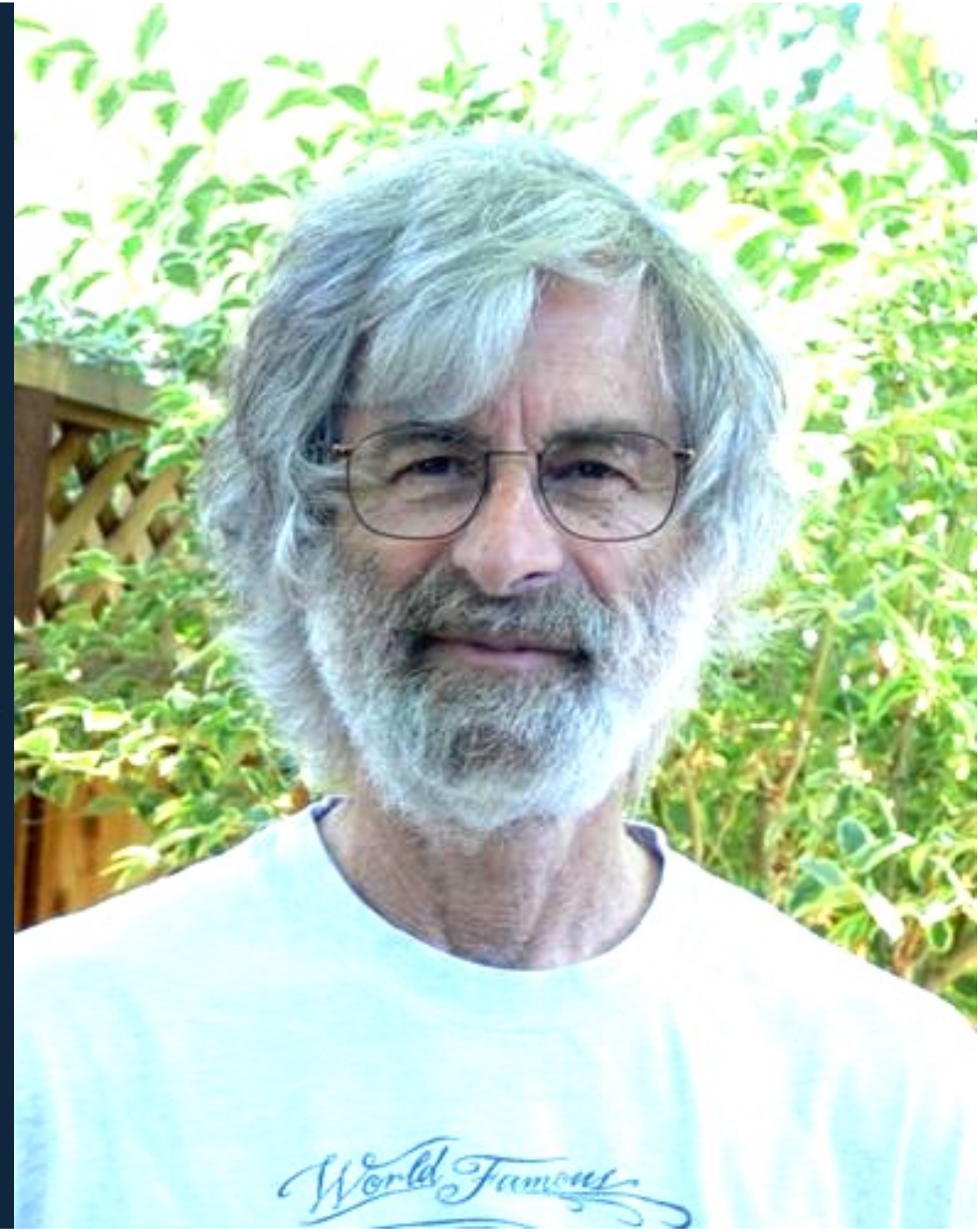
Medical Image Computing (MIC) CS-736

- **4 Assignments** (around 25% weight)
 - Implement algorithms in carefully curated (not wild) settings to learn concepts
 - Use any coding environment: Python, Matlab, C++, [Insight Toolkit \(ITK\)](#)
 - Experiment with real-world and simulated image data
- **1 Course Project** (around 20% weight)
 - Students choose topic: can meet instructor to discuss
 - Must implement top-tier research publication and reproduce its results
- **2 Exams** (around 50% weight)
 - Mid-semester exam (around 25% weight)
 - End-semester exam (around 25% weight)
- **Attendance** (around 5% weight)
- “around” means $\pm 10\%$ allowed, e.g., 30% to 20%, or 50% to 60%

Medical Image Computing (MIC) CS-736

"If you're thinking without writing, you only think you're thinking. [...] Everyone thinks they think. If you don't write down your thoughts, you are fooling yourself."

— Leslie Lamport, creator of LaTeX (a document preparation system)



Medical Image Computing (MIC) CS-736

- **Assignments**
- For maximum benefit, assignments must be done alone
 - Ask TAs, or instructor, about technical queries
- If done in a group, each student in the group must solve each and every question, independently or jointly
 - e.g., cannot split work as:
student A solves questions 1 and 2
student B solves questions 3 and 4
 - This is important for effective learning

Medical Image

- Plagiarism



Medical Image Computing (MIC) CS-736

- **Plagiarism** policy
 - Institute policy www.iitb.ac.in/newacadhome/punishments201521July.pdf
 - “A student found copying in an assignment/laboratory project is given a zero in the assignment/project and is further given a one grade penalty.”
 - “The same disciplinary action is taken against both the person copying and the person from whom the material was copied.”
- CSE policy
 - Teaching assistant or instructor will report case to [Departmental Academic Disciplinary Action Committee \(DADAC\)](#)
 - At least loss of 1 grade level
 - Can get a fail grade

Medical Image Computing (MIC) CS-736

- Plagiarism policy
 - <https://integrity.mit.edu/handbook/writing-code>
 - “..., during the time that you are helping another student, your own solution should not be visible, either to you or to them. Make a habit of closing your laptop while you’re helping.”
 - “As they type lines of code, they speak the code aloud to the other person, to make sure both people have the right code.
INAPPROPRIATE.”
 - “In a tricky part of the problem set, Alyssa and Ben look at each other’s screens and compare them so that they can get their code right.
INAPPROPRIATE.”

Medical Image Computing (MIC) CS-736

- Plagiarism policy <https://integrity.mit.edu/handbook/writing-code>
 - “Jerry opens his own laptop, finds his solution to the problem set, and refers to it while he’s helping Ben correct his code.
INAPPROPRIATE.”
 - “Ben has by now spent a couple hours with Louis, and Louis still needs help, but Ben really needs to get back to his own work. He puts his code in a Dropbox and shares it with Louis, after Louis promises only to look at it when he really has to.
INAPPROPRIATE.”
 - “They exchange their test cases with each other to check their work.
INAPPROPRIATE.
Test cases are part of the material for the problem set, and part of the learning experience of the course. You are copying if you use somebody else’s test cases, even if temporarily.”

Medical Image Computing (MIC) CS-736

- Matlab Tutorial 1
 - www.mccormick.northwestern.edu/documents/students/undergraduate/introduction-to-matlab.pdf
- Matlab Tutorial 2
 - web.eecs.umich.edu/~aey/eecs451/matlab.pdf
- Matlab Tutorial 3
 - www.cs.cmu.edu/~ggordon/10601/recitations/matlab/pretty_matlab_pres.pdf
- Matlab Image-Processing Toolbox Tutorial
 - www.cs.otago.ac.nz/cosc451/Resources/matlab_ipt_tutorial.pdf
- Writing fast matlab code
 - www.csc.kth.se/utbildning/kth/kurser/DN2255/ndiff13/matopt.pdf
- Matlab Tips and Tricks
 - www.ee.columbia.edu/~marios/matlab/mtt.pdf

Medical Image Computing (MIC) CS-736

- **Attendance** policy
- acad.iitb.ac.in/files/UG_RULE_BOOK.pdf
- "IIT Bombay expects one hundred percent attendance (100%) from its students in all classes."
- "If the attendance of the student, as counted with effect from the first contact hour, falls below eighty percent of the total attendance expected, the 'DX' grade will be awarded in that course."
- "Only exception to this rule are courses where the instructor has declared that no DX grade will be awarded."
- Medical certificate required from IITB hospital, or one duly authenticated by IITB Hospital.

Medical Image Computing (MIC) CS-736

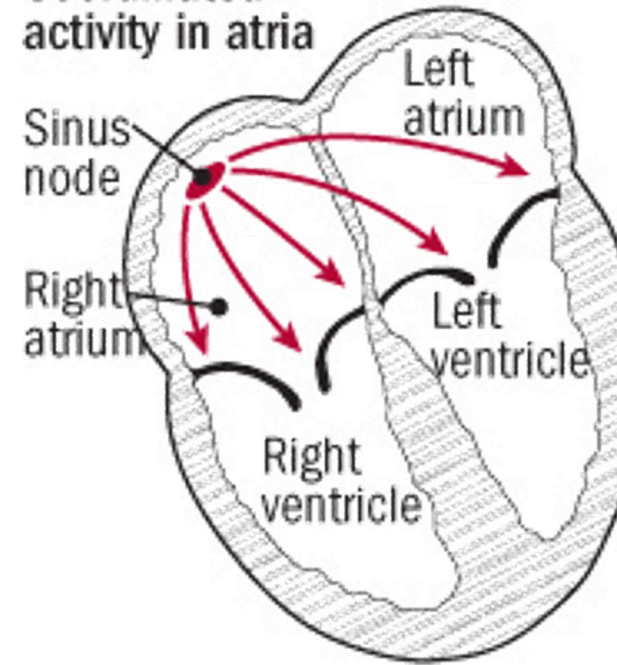
- Background
 - Necessary background = topics covered in IITB-CSE UG courses like:
 - Linear algebra. Probability and statistics. Data structures and algorithms. Programming.
 - Sufficient background = some course on:
 - Signal / image / audio / text processing
 - That covers the required basics
 - Machine learning and related courses
- Who can take course for credit ?
 - IITB PG students (with appropriate background)
 - IITB UG students from year-3 onwards (with appropriate background)
 - IITB-CSE BTech2 students: need AB or better in CS-215 & CS-213 & CS-293
- Who can take course for audit ?
 - Nobody

Signals, Images

- Function / signal
- Types of domains
 - Time
 - **Structured grid**
(e.g., regular grid in space) → Image
 - **Structured/regular grid in domain + Time** → “Dynamic” image
- Types of ranges
 - Integer, real
 - Scalar, vector
- Dimensions
 - $1 \rightarrow 1$: ECG
 - $2 \rightarrow 1$: Image (grayscale)
 - $3 \rightarrow 3$: Video (color; no sound)

A. Normal sinus rhythm

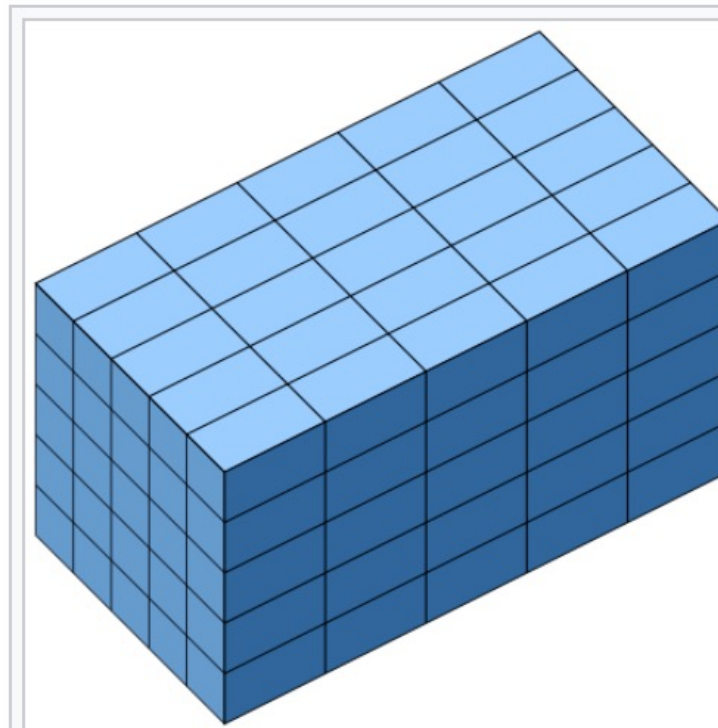
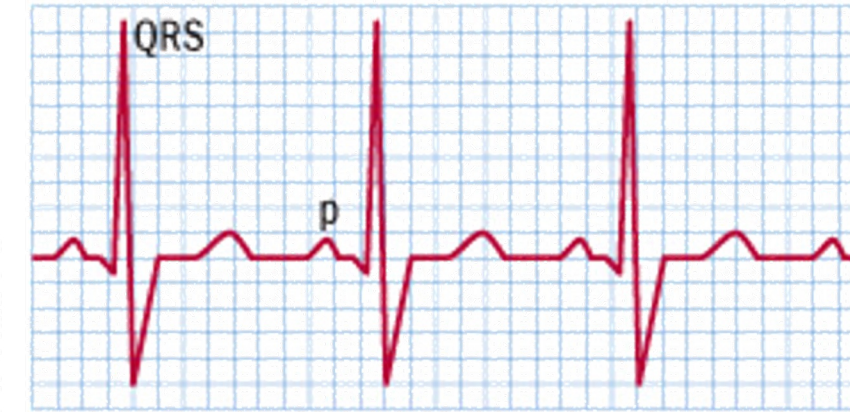
Coordinated activity in atria



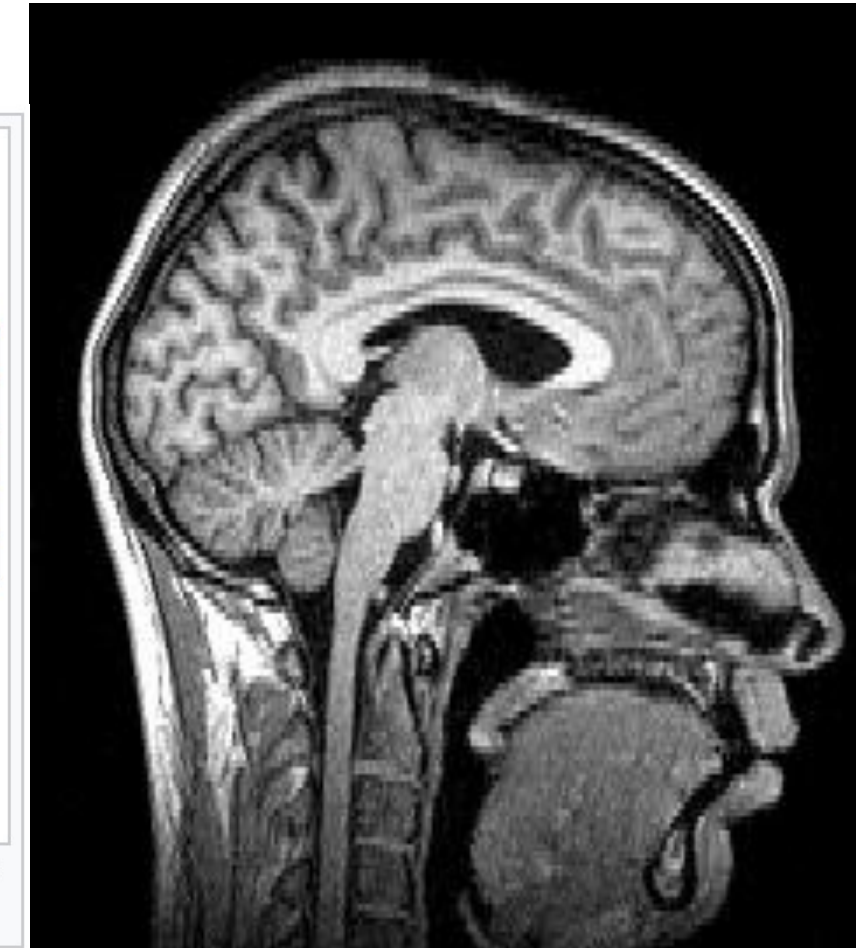
Electrocardiogram (ECG)

25 squares = 1 second

Normal sinus rhythm



Example of a regular grid



Signals, Images

- This course **isn't** about medical (non-image) **signals**

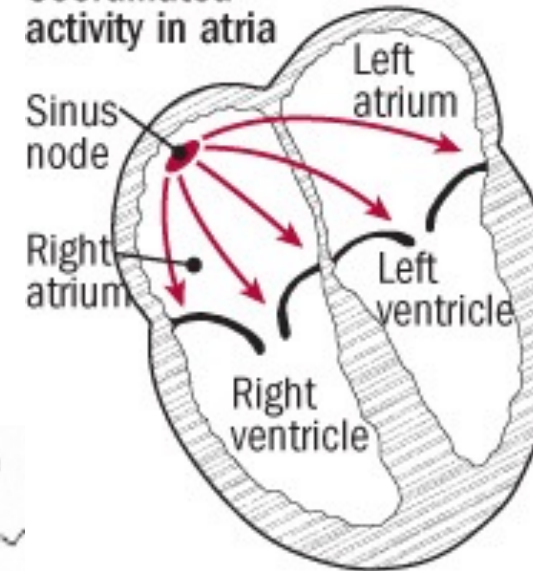
- Electro-cardio-gram (ECG)

- Electro-encephalo-gram (EEG)

- Domain
= unstructured grid
in space

A. Normal sinus rhythm

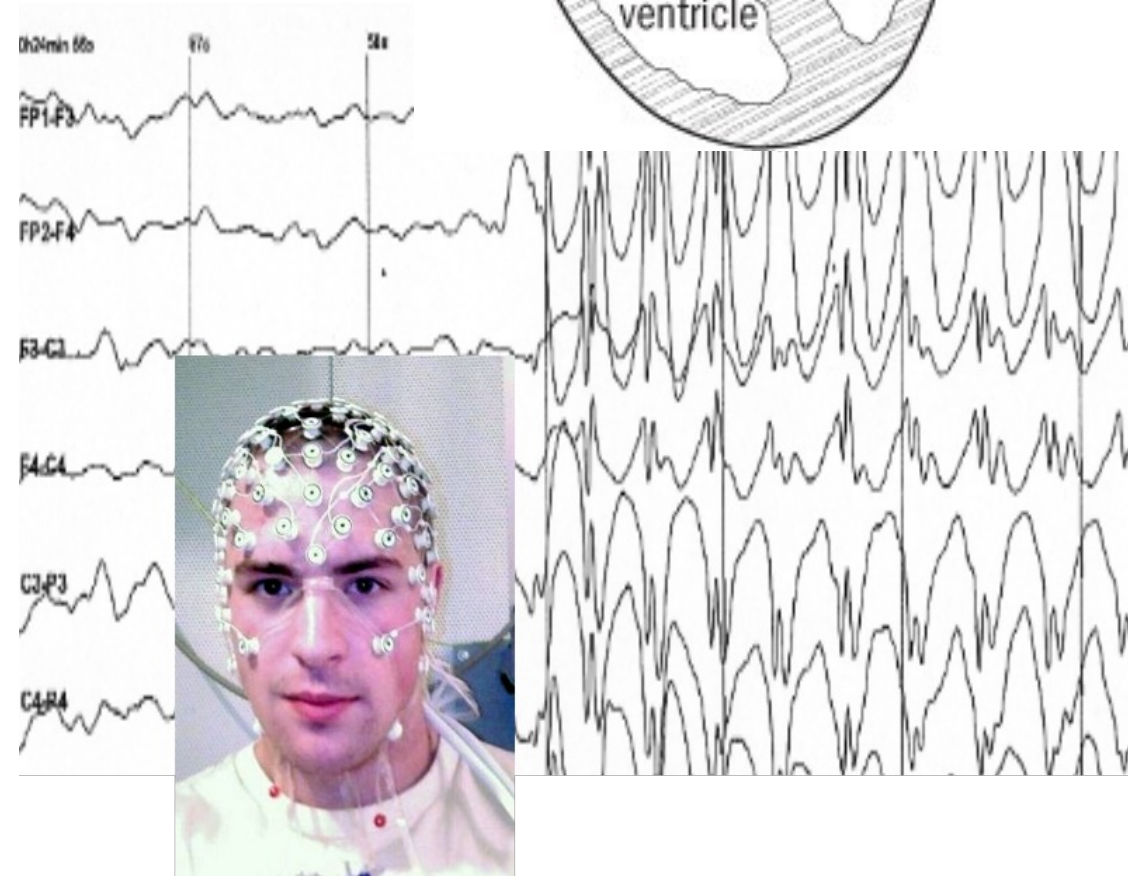
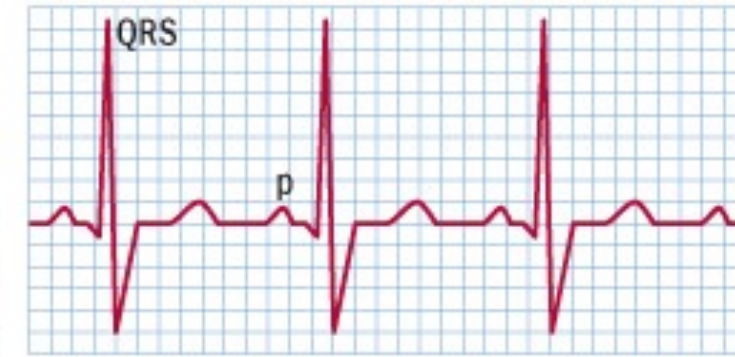
Coordinated activity in atria



Electrocardiogram (ECG)

25 squares = 1 second

Normal sinus rhythm



- This course **is** about medical **images**

- X-ray, CT, MRI, ultrasound, microscopy, etc.

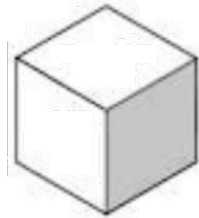
Analog, Digital

- “Analog” Image
 - Function (data) on a “continuous” domain
 - No discretization or quantization
 - Data acquired on photographic film (“negative”), e.g., X-ray film
 - X-rays on films
 - Microscopy on films
 - Resolution only limited by film-grain size
- This course **isn't** about **analog** images
 - Digital images
 - Digital X-ray scan
 - Digital microscopy
 - CT, MRI, ultrasound
 - Acquisition is digital by scanner-system design

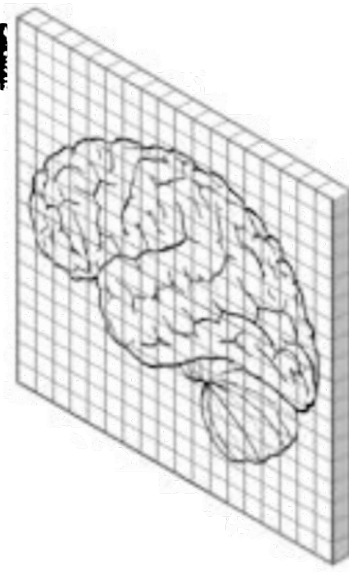


Analog, Digital

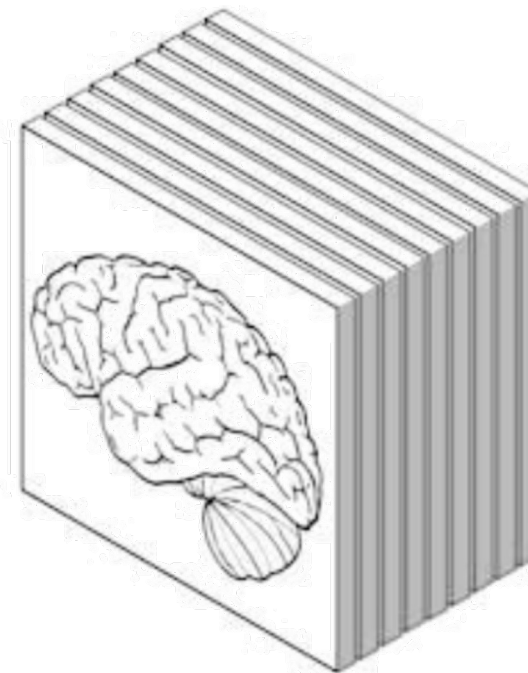
- **Digital image:**
Array of numbers (one/multi dimensional)
 - Integers, real
 - Signed, unsigned (non negative)
- For 2D Cartesian spatial grid
 - “pix” (picture) + “el” (element) = “pixel”
 - Each pixel assigned a value / intensity
- For 3D Cartesian spatial grid
 - “volume element” = **voxel**
 - Each voxel assigned a value / intensity



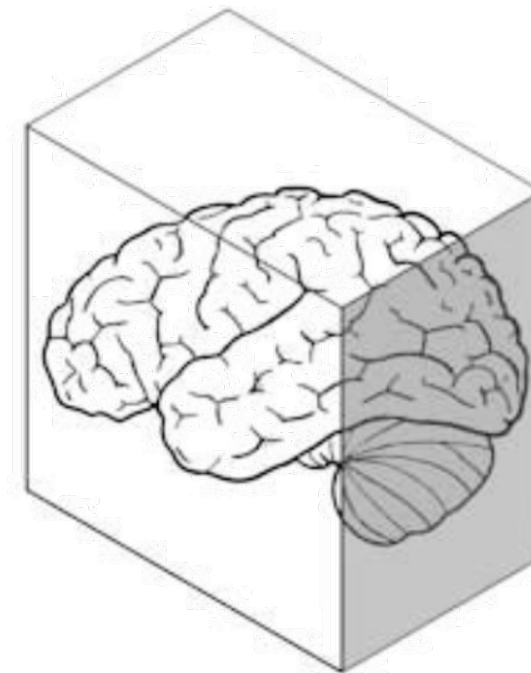
VOXEL



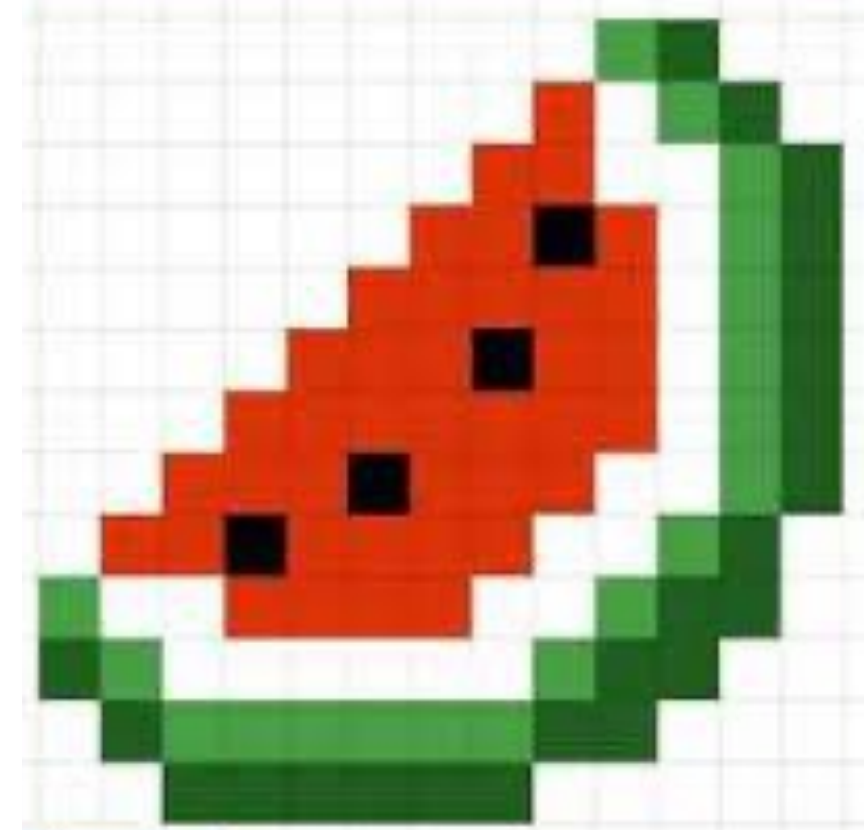
INDIVIDUAL SLICE



SLICES

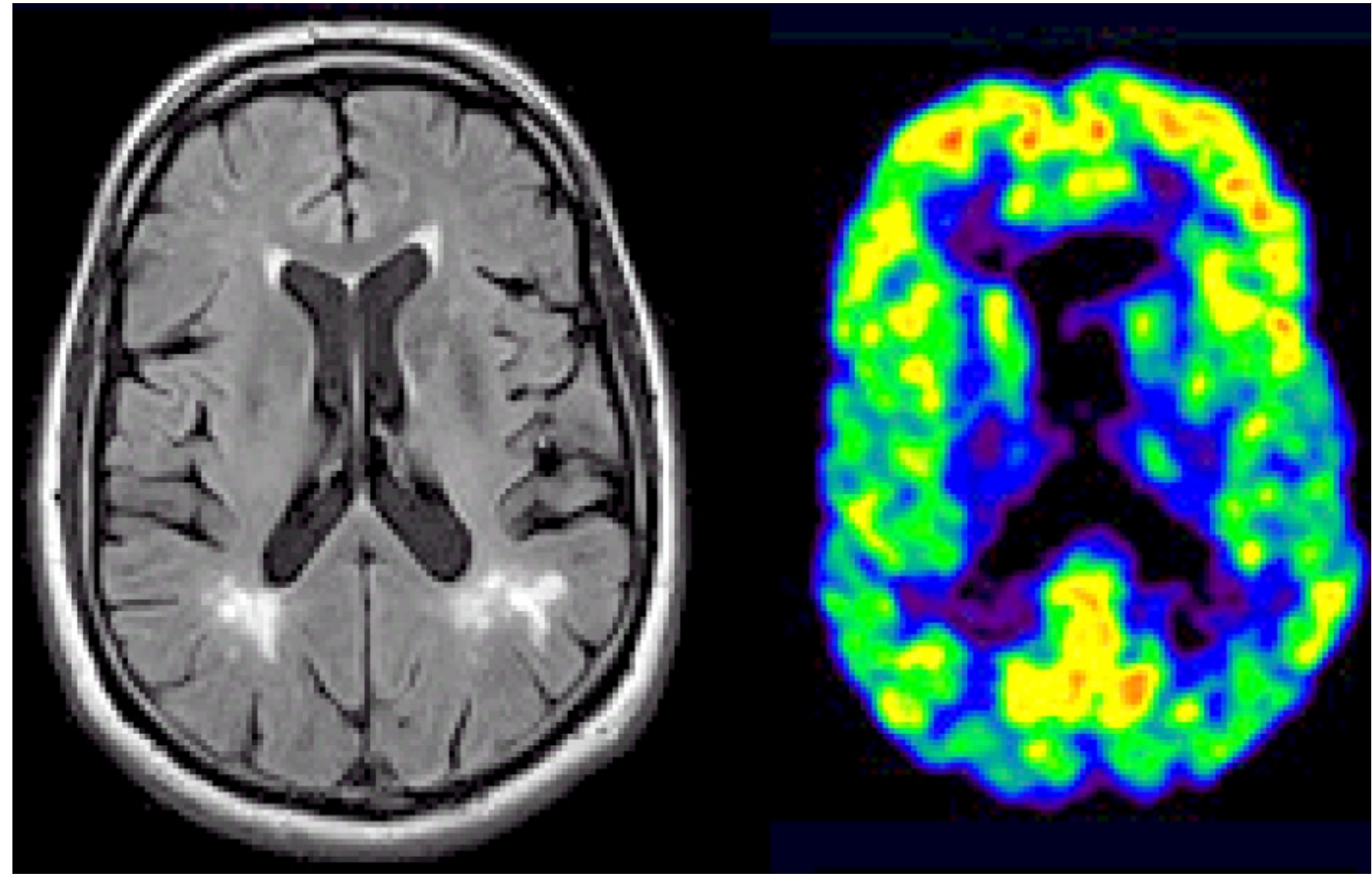


VOLUME



Digital Image

- Colormap:
Mapping from signal/data values to colors (for visualization)
- Used purely for display purposes
- Examples
 - Left: MRI
 - Data = matrix of scalars
 - Visualization: grayscale
 - Right: PET
 - Data = matrix of scalars
 - Visualization: colored heat-map



Imaging

- **Imaging = Process of acquiring image data**
 - Taking a digital photograph
 - Taking an X-ray/CT/MRI scan/measurements
 - May or may not produce a clinically interpretable image
- 1) Imaging Hardware / Instrumentation
 - Camera lens system in microscopes, X-ray machine instrumentation, CT ..., MRI ..., etc.
- 2) Imaging Physics
 - Microscopy → Optics (visible spectrum)
 - MRI → Electromagnetism (lower / radio frequency)
 - CT → Electromagnetism (higher frequency, ionizing radiation)
 - Ultrasound → Acoustics
- This course **isn't** about **imaging**
 - We don't study how to build a machine, or all details of underlying physics

Imaging

- This course **is** about **image-data processing (using computations) and analysis**
- Image data is given
- Scanner is a black box
 - Don't know all hardware / instrumentation details
 - e.g., what is inside machines
 - Don't know all physics details
 - e.g., optics / electromagnetism / acoustics
 - But we know high-level / mathematical system specification
 - Transformation from physical quantity $\rightarrow$ data
 - e.g., colors and objects in scene $\rightarrow$ values in acquired image
 - e.g., tissues in body $\rightarrow$ acquired data

Medical Image Computing

- **Applications** are usually of 2 kinds

1) Clinical

- Diagnosis
 - e.g., dental cavity, fracture, lung disease, cancer, dementia, fetus
- Interventional
 - e.g., image-guided catheter to remove blockages in artery
 - e.g., surgical planning
- Therapy
 - e.g., image-guided radiation therapy
- Monitoring
 - e.g., tumor reduction over time

Medical Image Computing

- **Applications** are usually of 2 kinds

2) Scientific exploration / understanding

- Study effects of pathology on *structure (anatomy)*
 - e.g., dementia (brain), cardiac arrhythmia (heart)
 - e.g., size, shape, location, correlation with clinical variable
- Study effects of pathology on *function (physiology)*
 - e.g., brain function, heart muscle function
 - e.g., is tissue dead ? is activity / blood flow reduced ?
- Study effects of drug on pathology, structure, function
 - e.g., chemotherapy on cancer
 - Also study side effects, toxicology

Medical Image Computing



Medical Image Computing

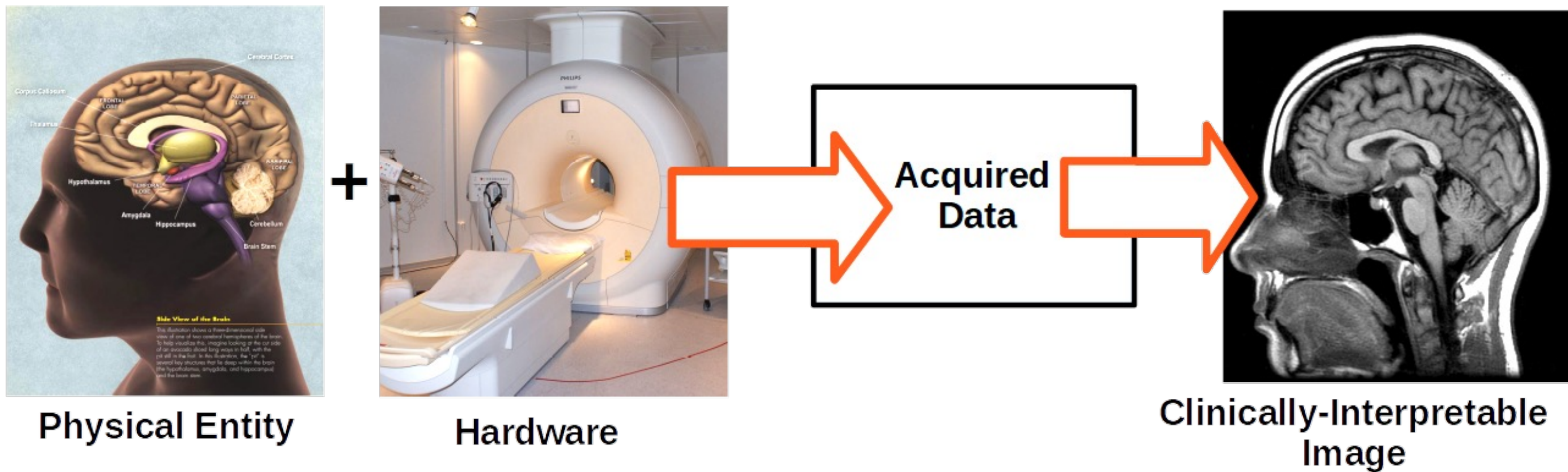
- Applications: Various kinds of processing
- Large set of subjects (reproducible, scalable)
- Improve imaging
 - Faster, lower radiation, different information
- Measure
 - Volume, area, thickness, shape: measurements over time
- Learn variation in structure / function in normal cases
 - Help diagnose / detect abnormality
- Classify
 - Help detect disease: disease-A1 / disease-A2 / disease-A3 / other
- Test hypothesis
 - Therapy restores structure/function, Disease effects size / shape of structure

Medical Image Computing: Example Applications

- All of these (in following slides) can lead to great course projects
- You can start thinking about it ...
- ... starting now !

Medical Image Computing: Example Applications

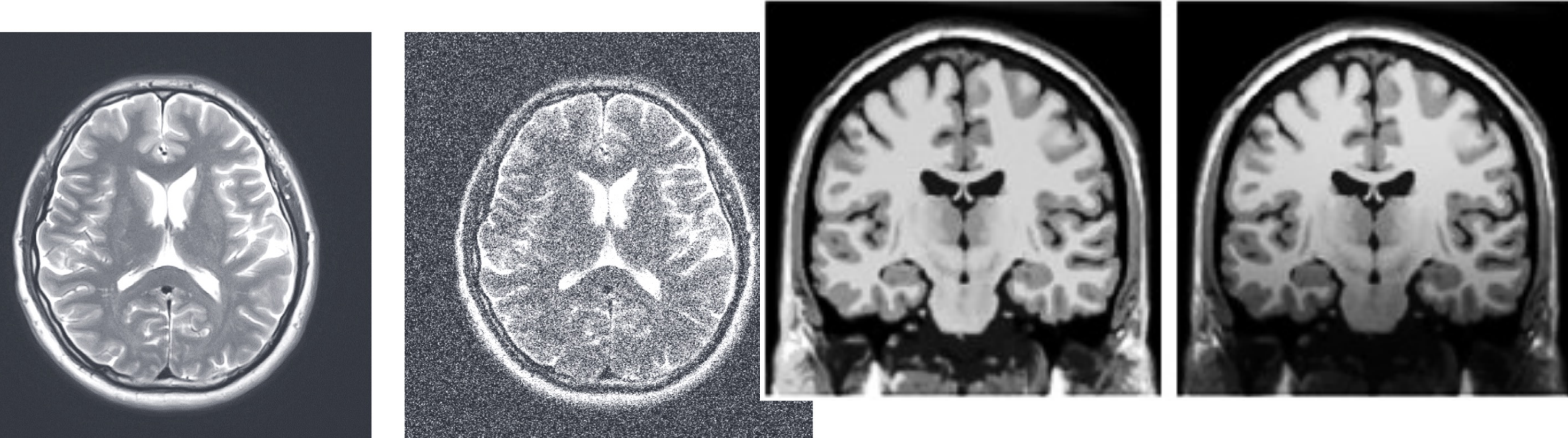
- **Application: Image Reconstruction**
- Generate an image from raw scanner data
 - Understand principles of data acquisition



- Refers to some basic concepts in image/signal processing
 - Convolution, filtering, Fourier representation & transform

Medical Image Computing: Example Applications

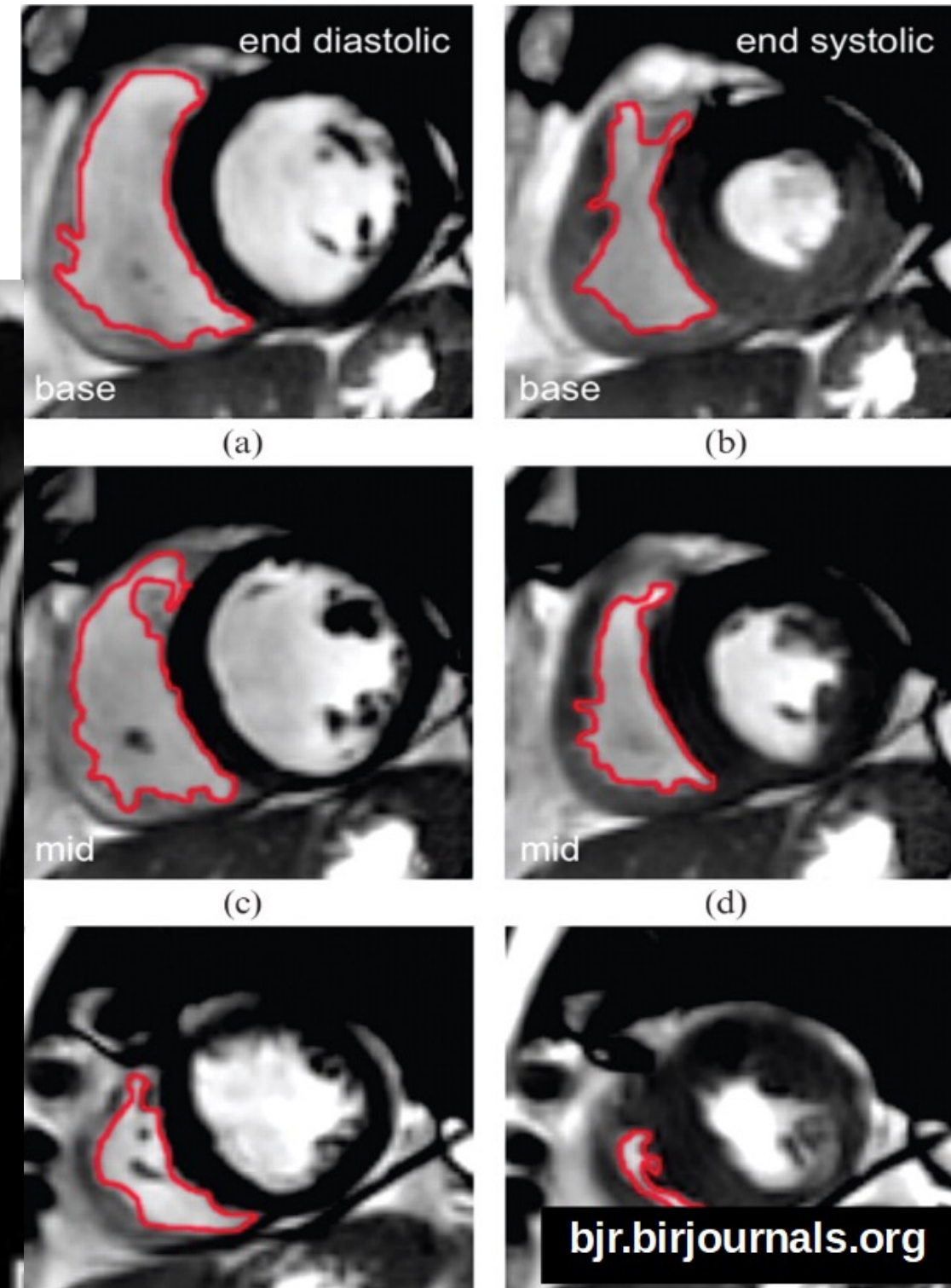
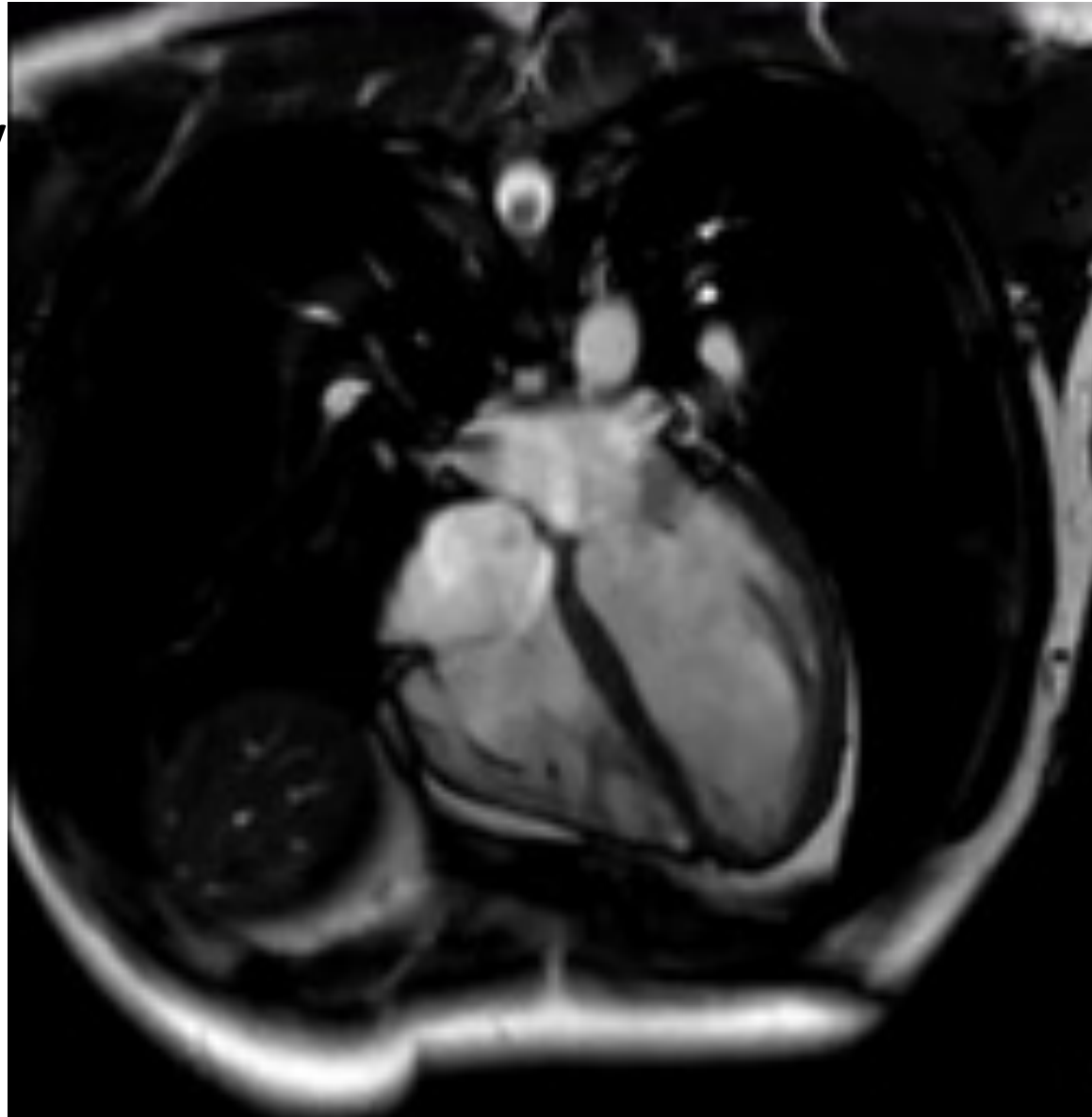
- **Application: Image De-noising / Restoration**
- Reducing noise / degradation
 - Understanding mathematical models for noise / corruption during imaging
 - Mathematical and statistical modeling, optimization
 - Modeling image, noise / corruption
 - How do algorithms exploit information from degradation model ?



Medical Image Computing: Example Applications

- **Application: Image Segmentation / Classification**

- Partitioning image into components / structures
- Delineating a structure within image
- Recognizing image class

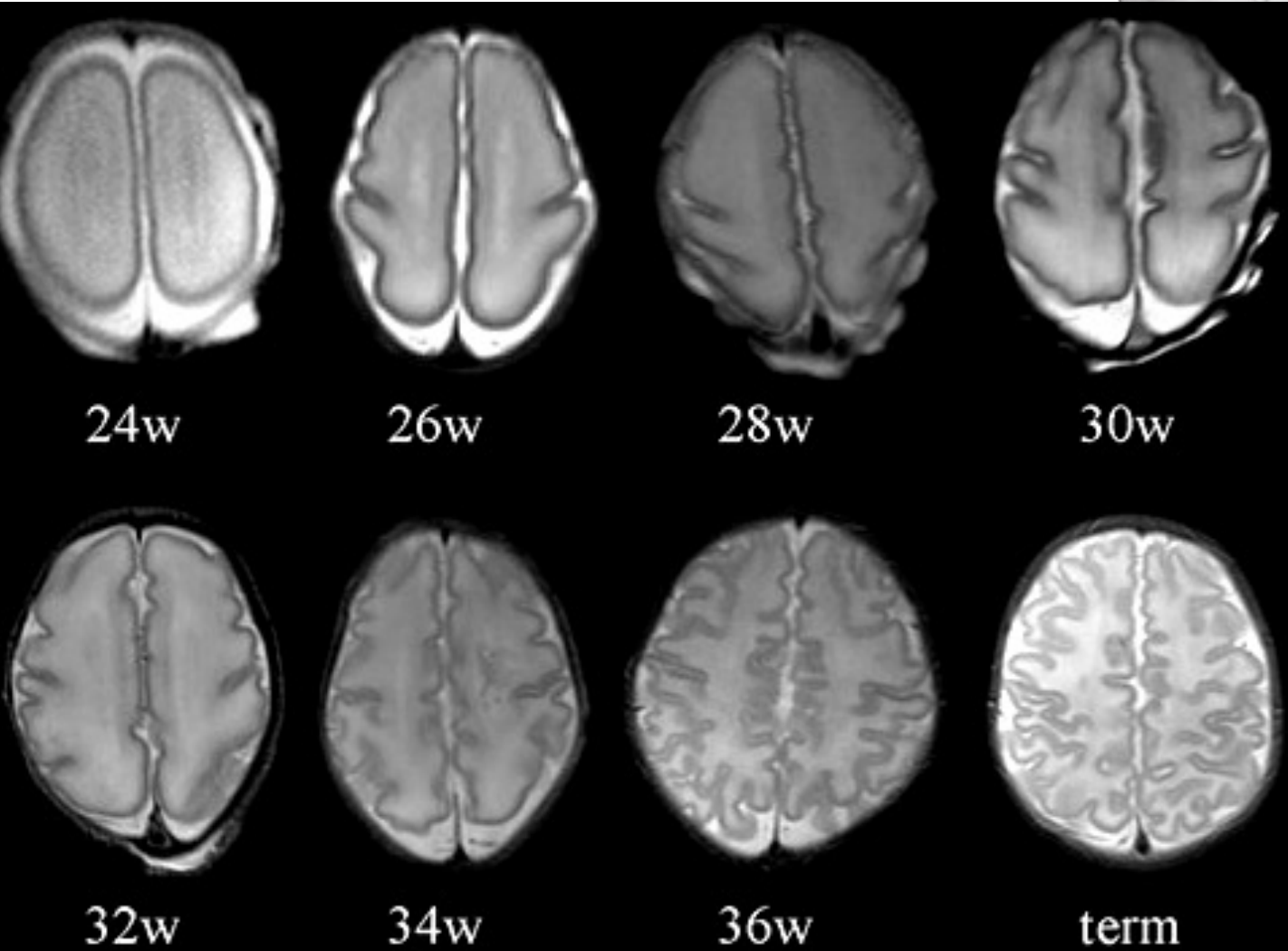
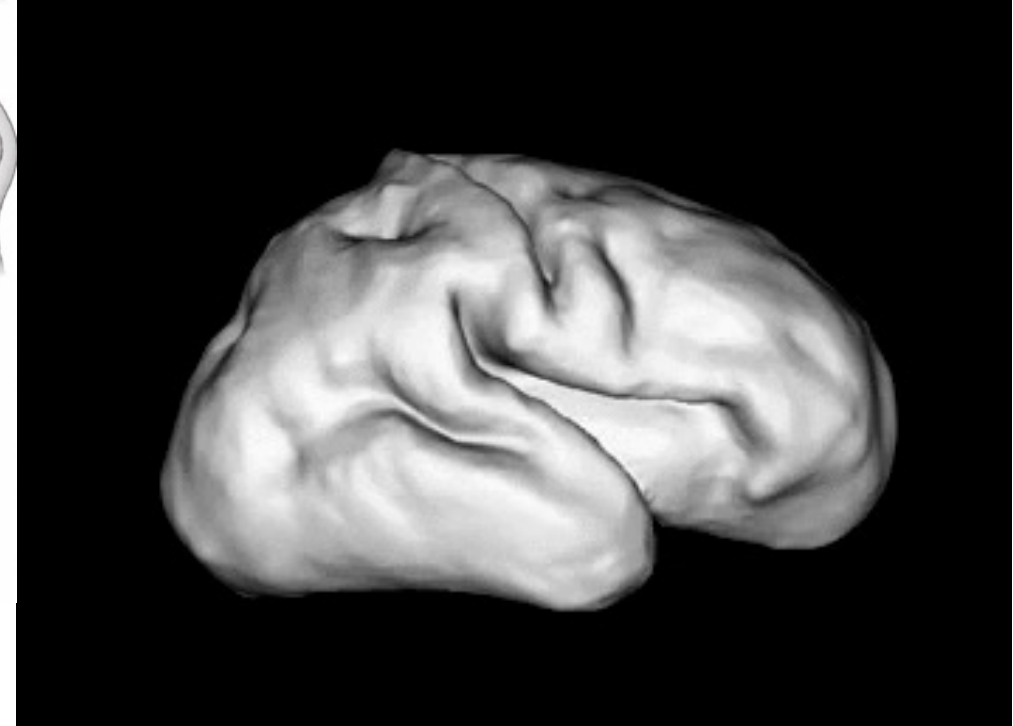
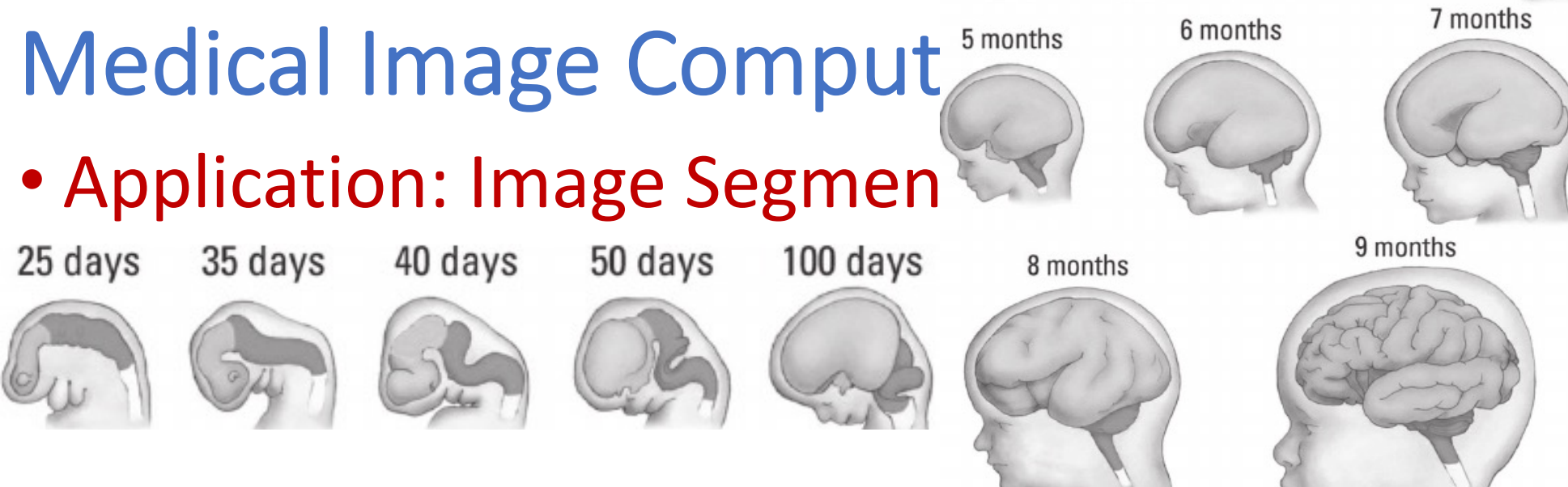


Medical Image Computing: Example Applications

- **Application: Image Segmentation**
- Principles in machine learning
 - Bayesian modeling, variational inference
 - Mathematical derivations from scratch
- Concepts in graph theory, discrete optimization
 - Mathematical derivations from scratch
- Concepts in deformable contours
 - Based on principles in partial differential equations

Medical Image Comput

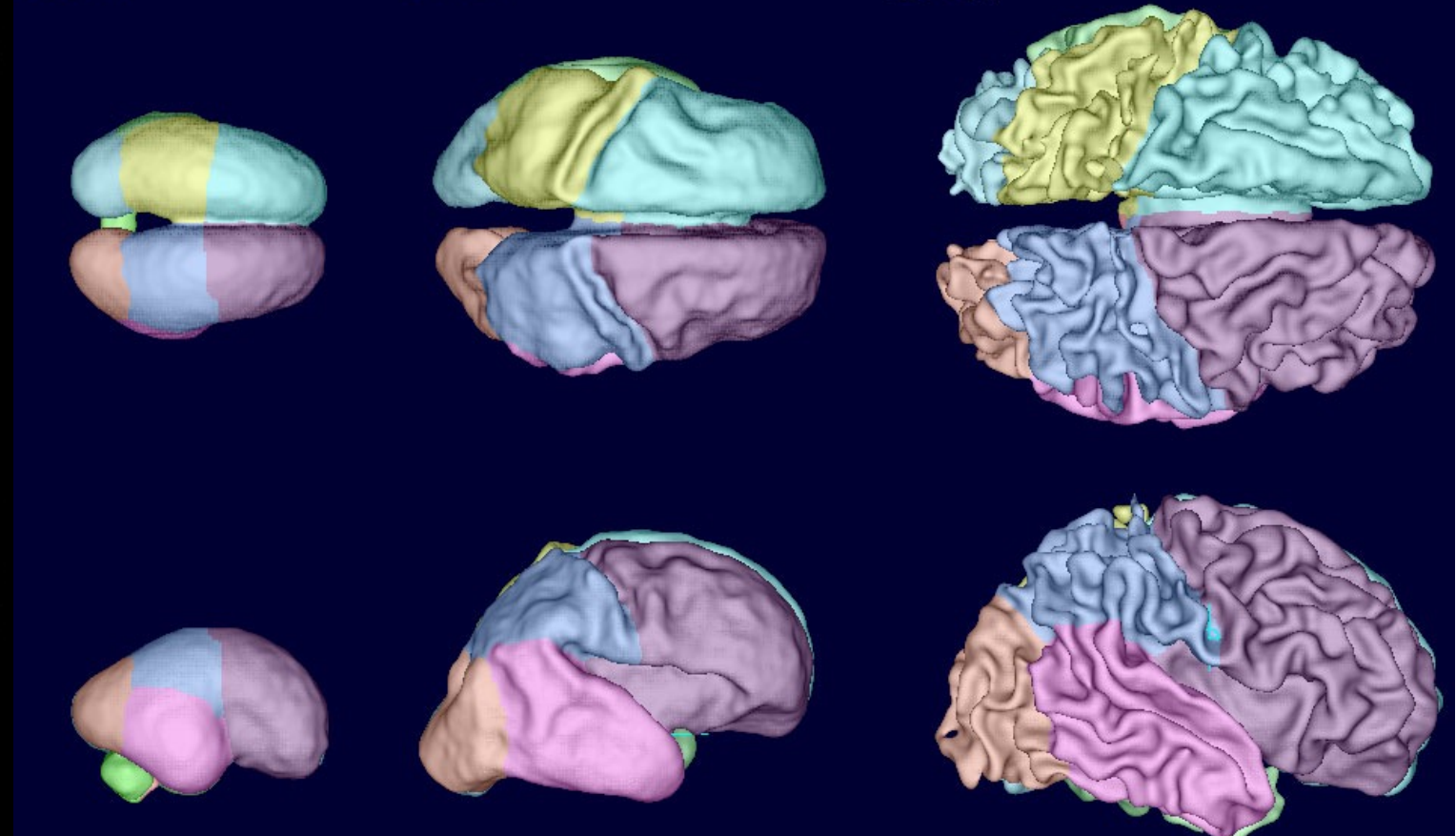
- Application: Image Segmen



20GW

29GW

35GW



Medical Image Computing: Example Applications

- **Application: Image Segmentation**
- Microsoft Research, Cambridge:
Tumor delineation using random forest

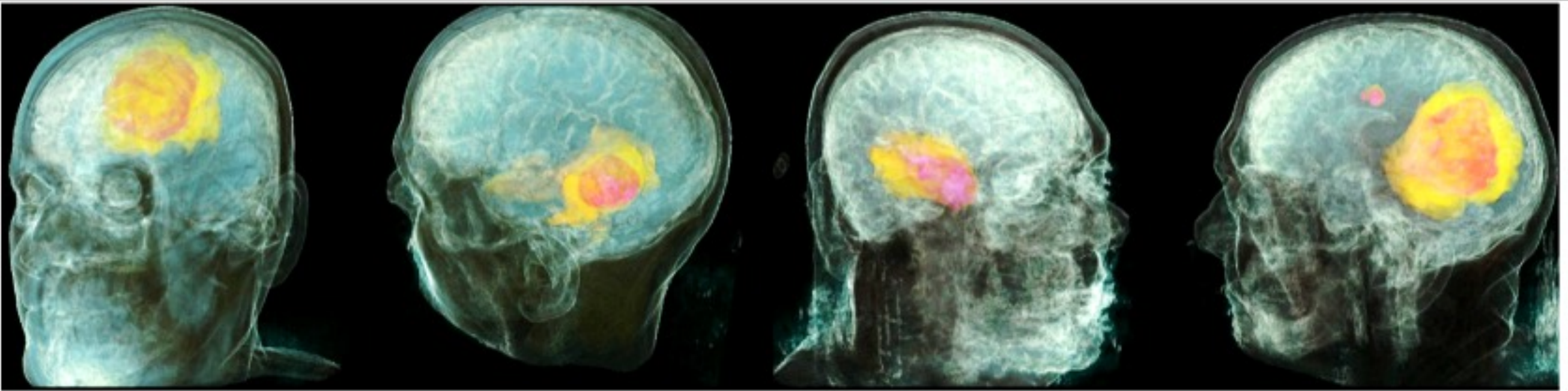
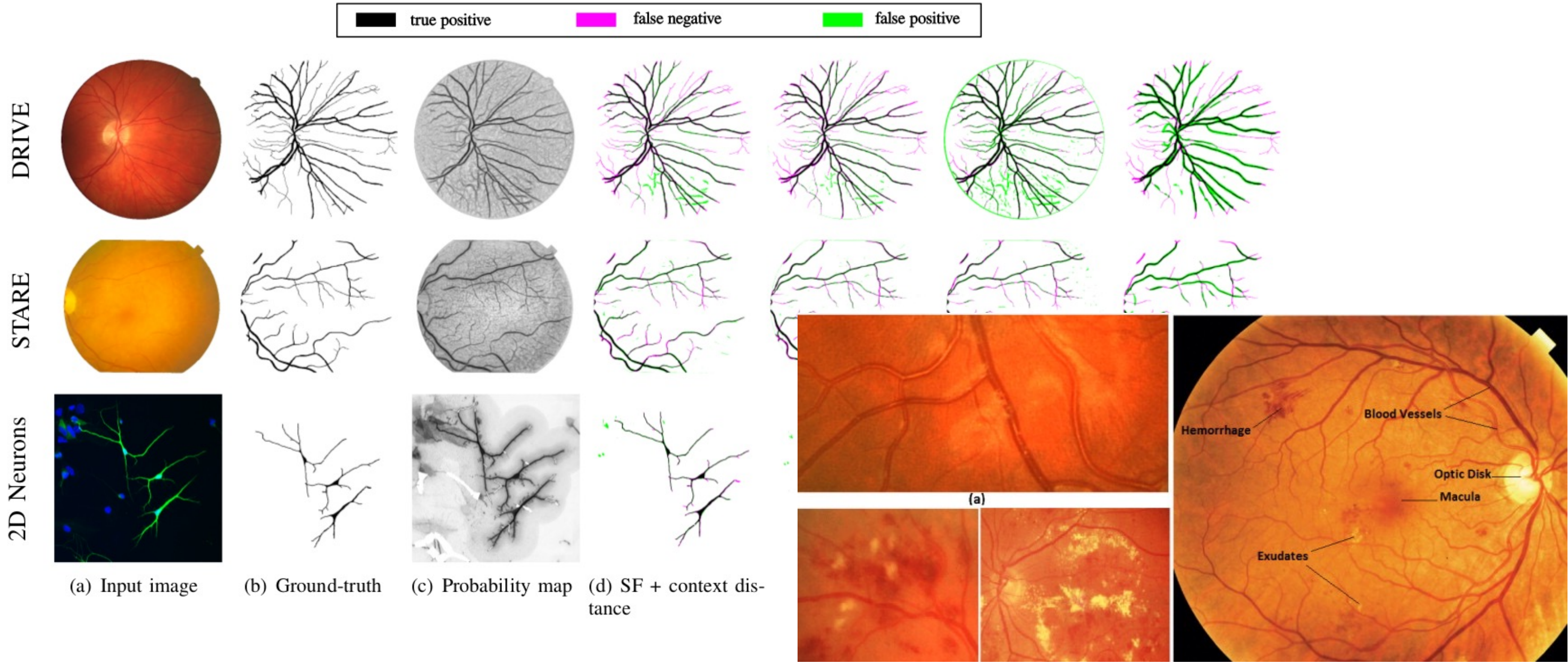


Fig. 3. Automatic 3D segmentation of glioblastoma and its constituent tissues for four different patients. Red indicates the active rim, yellow the oedema and green the necrotic core. Results obtained by running the forest in Zikic et al. (2012) on clinical MR.

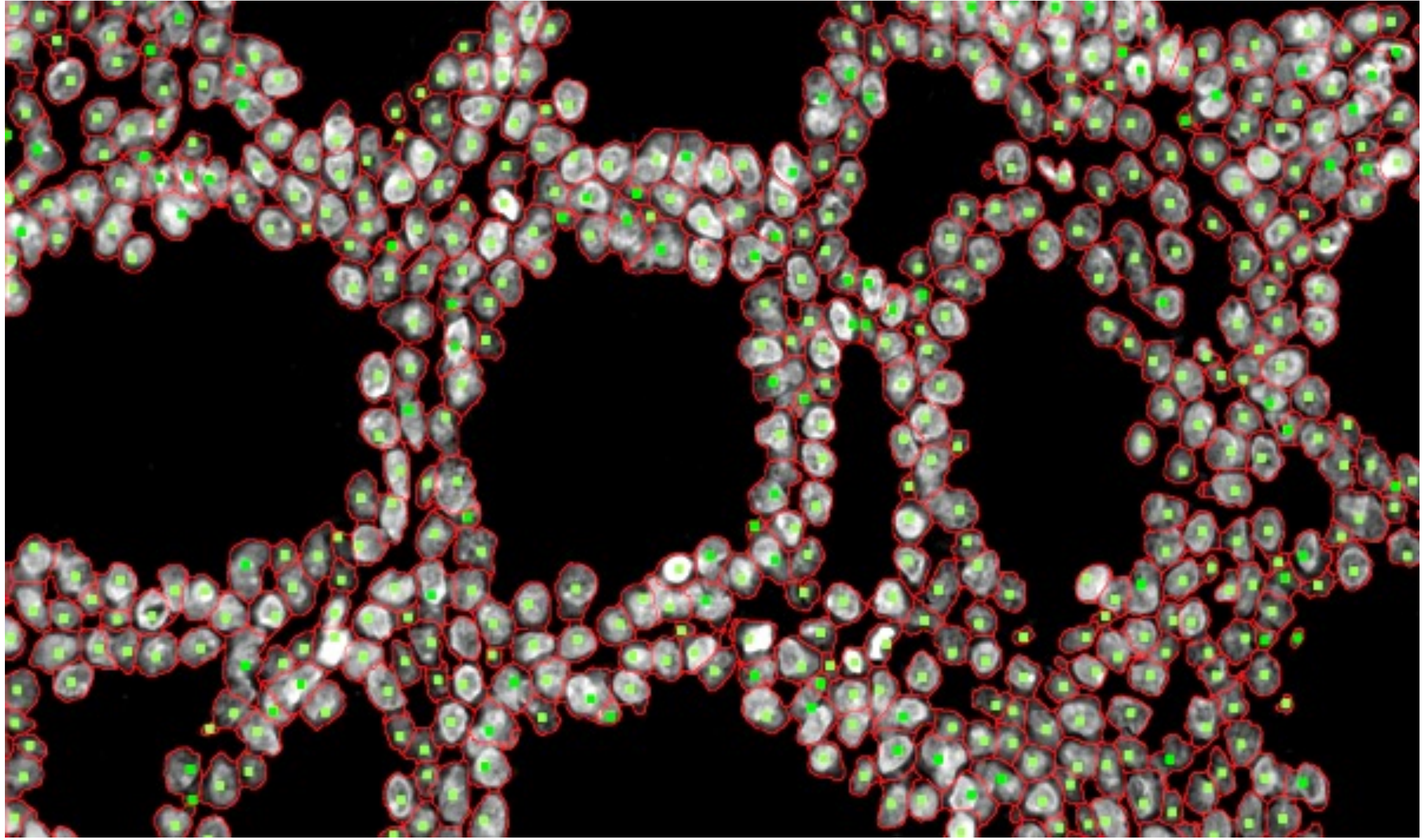
Medical Image Computing: Example Applications

- Application: Image Segmentation
- Blood vessels in retina



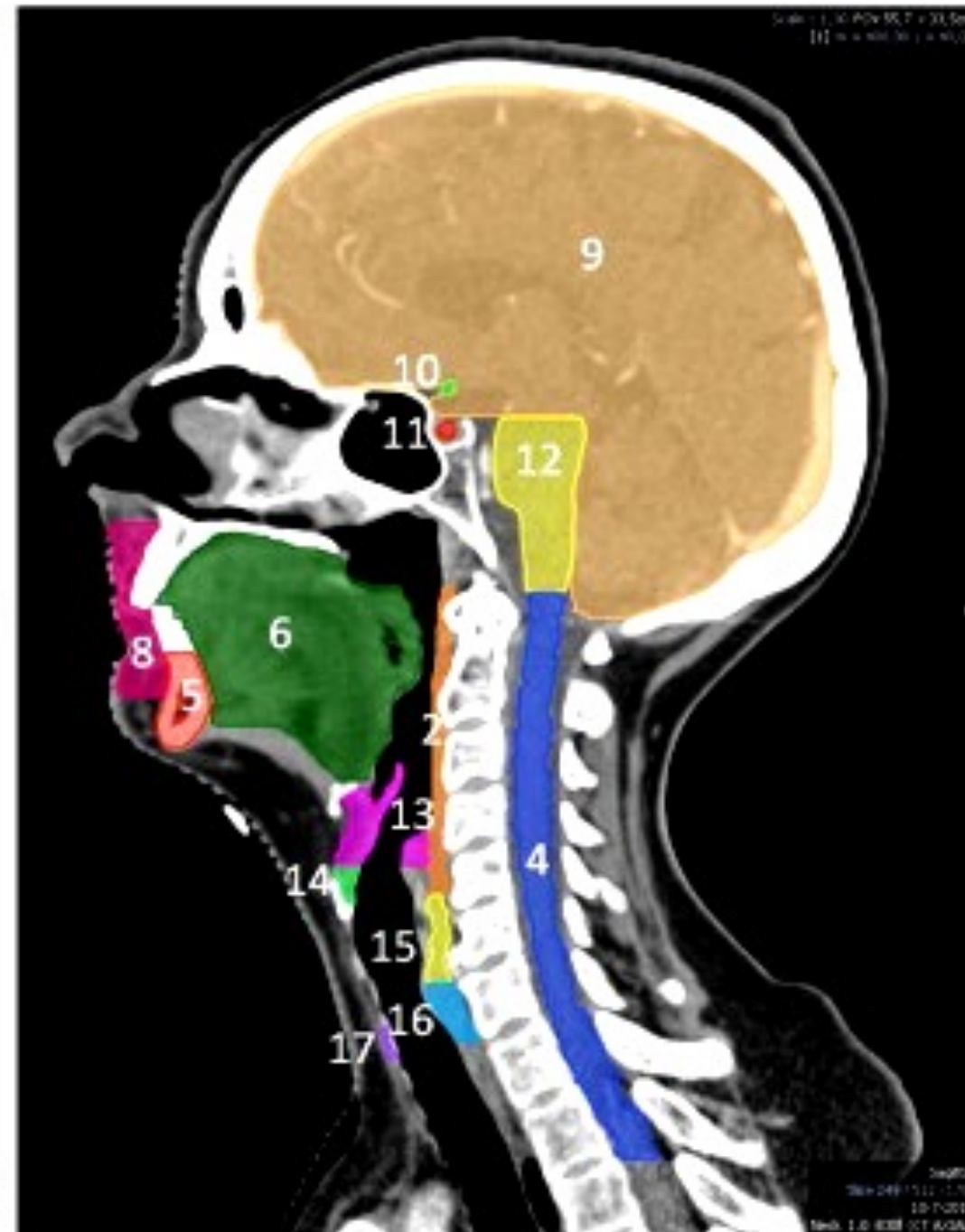
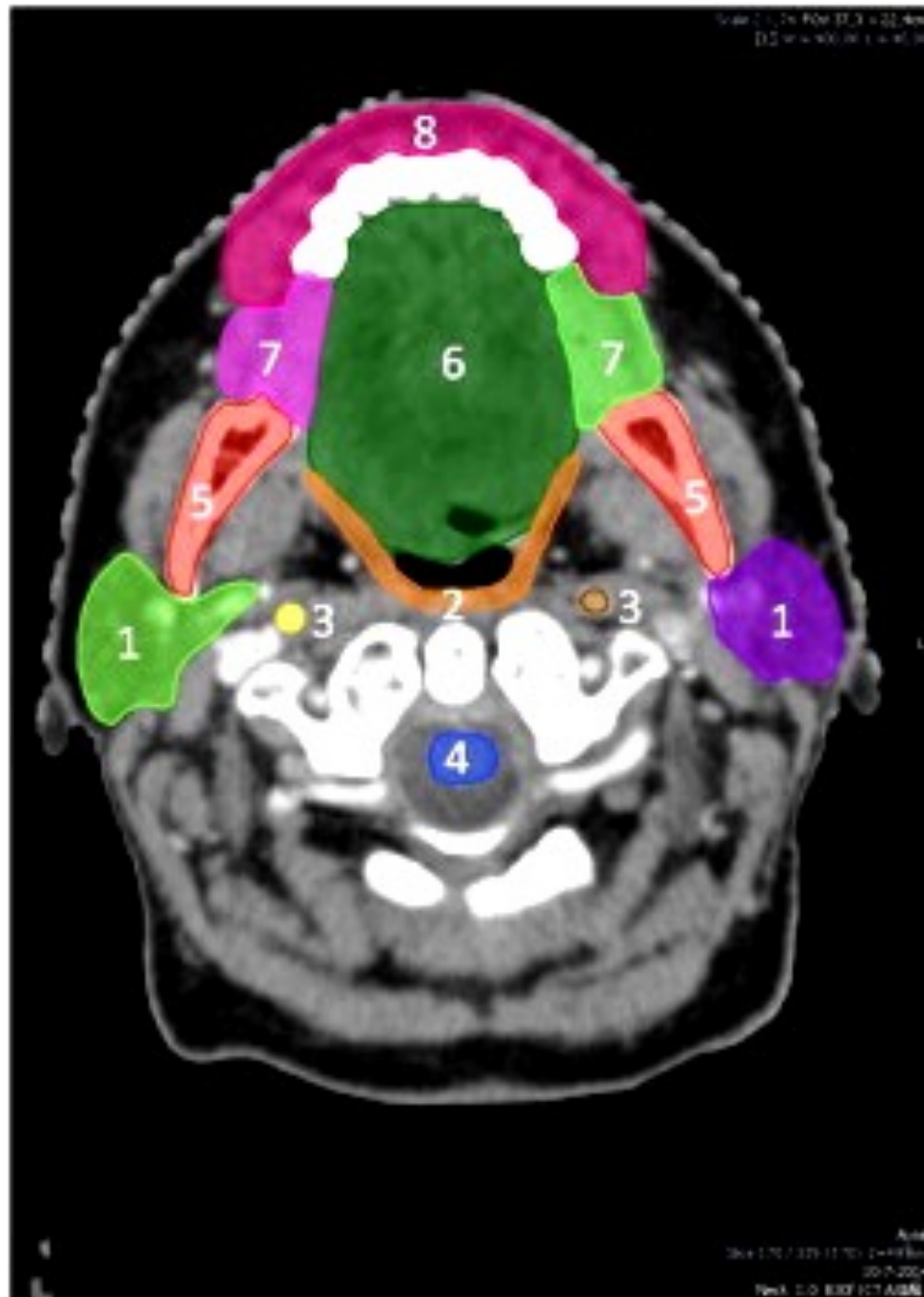
Medical Image Computing: Example Applications

- Application: Image Segmentation



Medical Image Computing: Example Applications

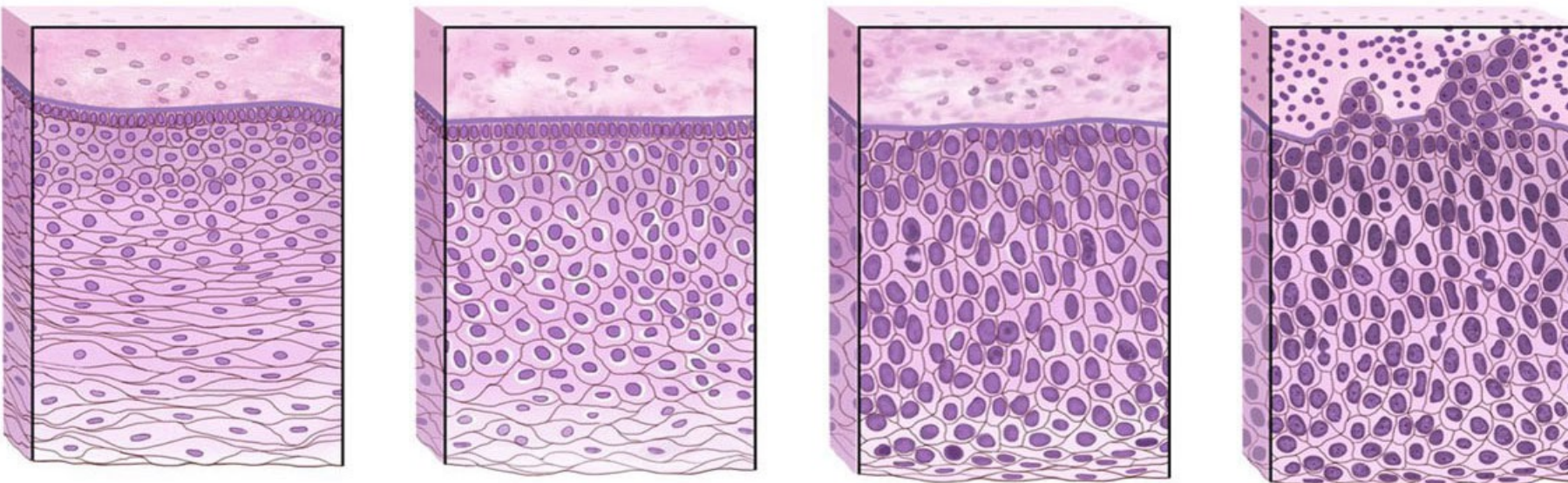
- Application: Image Segmentation



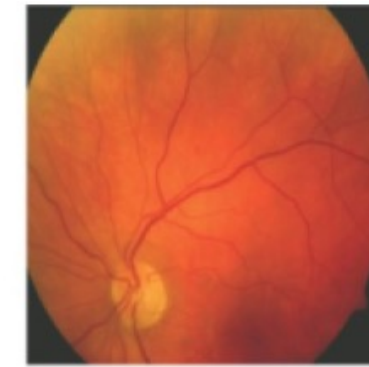
Medical Image Computing: Example Applications

- **Application: Image Classification**
- Classification
 - Automated algorithms can save time
 - Classifying (or regressing)
 - Disease type
 - Tumor type
 - Cancer grade (growth rate), stage (spread)

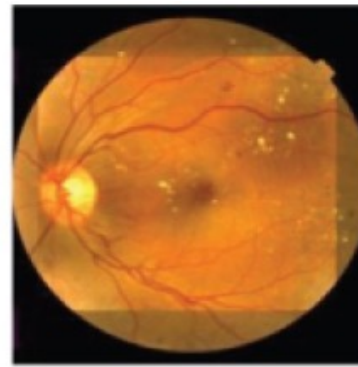
Normal → **Hyperplasia** → **Dysplasia** → **Cancer**



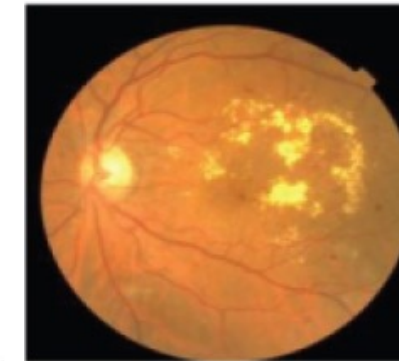
Without DR



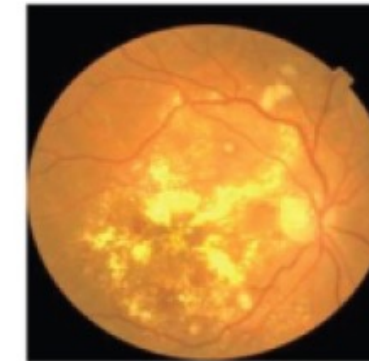
Early diabetic retinopathy



Mild NPDR



Moderate NPDR



Severe NPDR



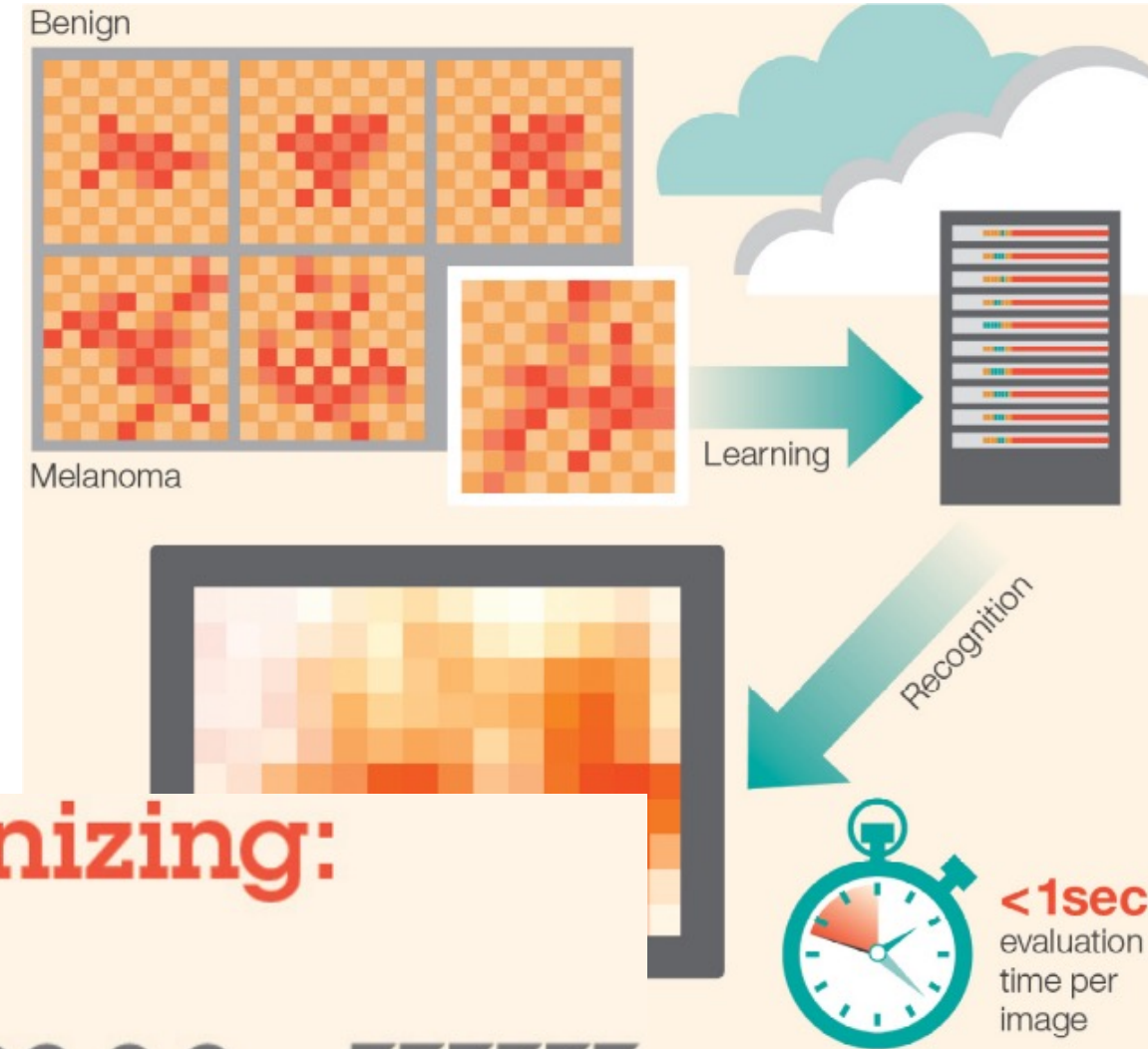
PDR and neovascularization

Medical Image Computing: Example Applications

- Application: Image Classification

IBM Research explores cognitive computing technology to help recognize skin cancer

As part of a collaborative effort with Memorial Sloan Kettering, IBM is researching the application of cognitive computing to analyze dermatological images of skin lesions with the goal of assisting clinicians in the identification of various cancerous disease states. The technology, which learns by identifying specific patterns in medical images, has the potential to increase the number of cases detected and help clinicians make earlier diagnoses.



What is a cognitive system recognizing:



color



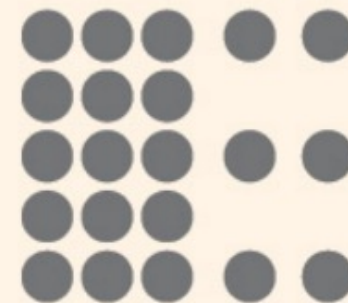
shape



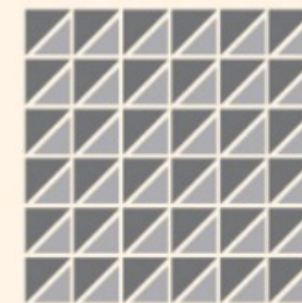
edges



pattern



density



texture

By images, technology to recognize diseased states +, compared to previously irrent clinical experts that are

Medical Image Computing: Example Applications

- Application: Image Classification

From Kinect to InnerEye – How Microsoft is supercharging gaming tech with AI smarts to help diagnose cancer

By Gerald Lynch May 09, 2017 World of tech

news.stanford.edu/2017/01/25/artificial-intelligence-used-identify-skin-cancer/

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JANUARY 25, 2017

Deep learning algorithm does as well as dermatologists in identifying skin cancer

In hopes of creating better access to medical care, Stanford researchers have trained an algorithm to diagnose skin cancer.

SCIENTIFIC REPORTS

OPEN

Deep learning as a tool for increased accuracy and efficiency of histopathological diagnosis

Received: 28 January 2016

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Geert Litjens¹, Clara I. Sánchez², Nadya Timofeeva¹, Meyke Hermesen¹, Iris Nagtegaal¹, Iringo Kovacs³, Christina Hulsbergen - van de Kaa¹, Peter Bult¹, Bram van Ginneken² & Jeroen van der Laak¹

Detecting Cancer Metastases on Gigapixel Pathology Images

Yun Liu^{1*}, Krishna Gadepalli¹, Mohammad Norouzi¹, George E. Dahl¹, Timo Kohlberger¹, Aleksey Boyko¹, Subhashini Venugopalan^{2**}, Aleksei Timofeev², Philip Q. Nelson², Greg S. Corrado¹, Jason D. Hipp³, Lily Peng¹, and Martin C. Stumpe¹

{liuyun,mnorouzi,gdahl,lhpeng,mstumpe}@google.com

¹Google Brain, ²Google Inc, ³Verily Life Sciences, Mountain View, CA, USA

Medical Image Computing: Example Applications

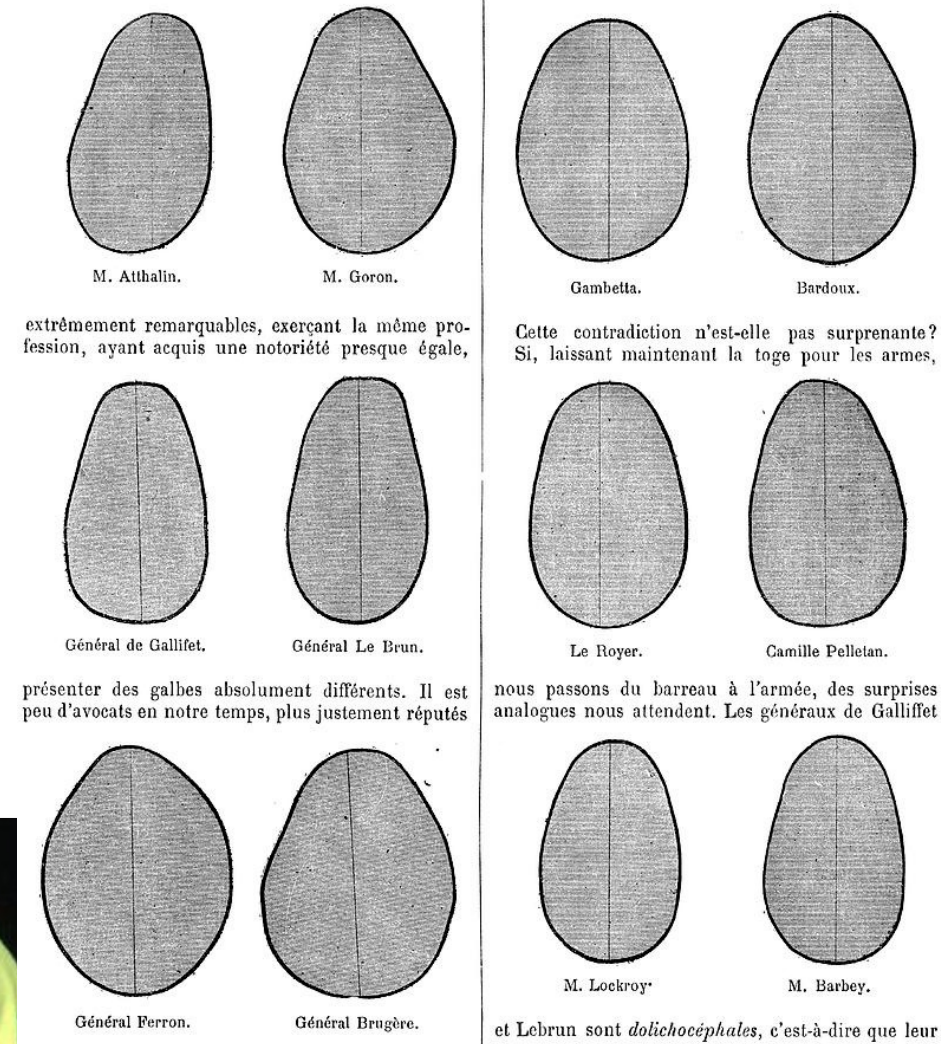
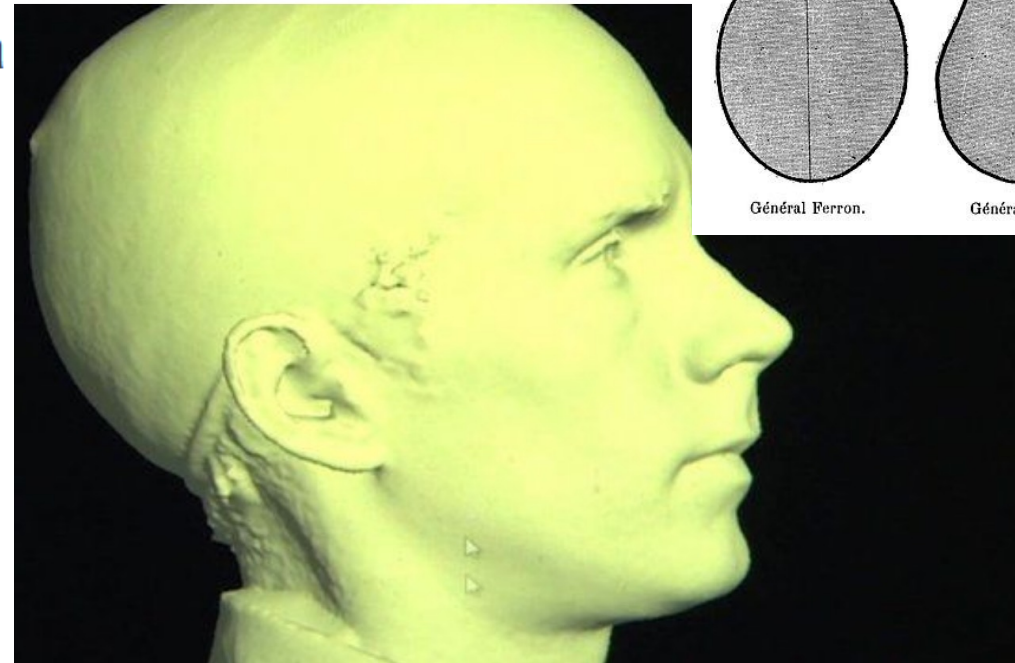
- **Application: Shape Analysis**
 - Normal / abnormal shape of a structure
- ## Defining the most 'normal' shape of a human skull

Reconstructive surgery for people born with a misshapen skull is one of medicine's most complicated challenges.

Part of the problem for surgeons is defining what a normal head actually looks like.

A team of Liverpool-based surgeons, and artists, have been hoping to find the answer.

Tim Muffett reports.

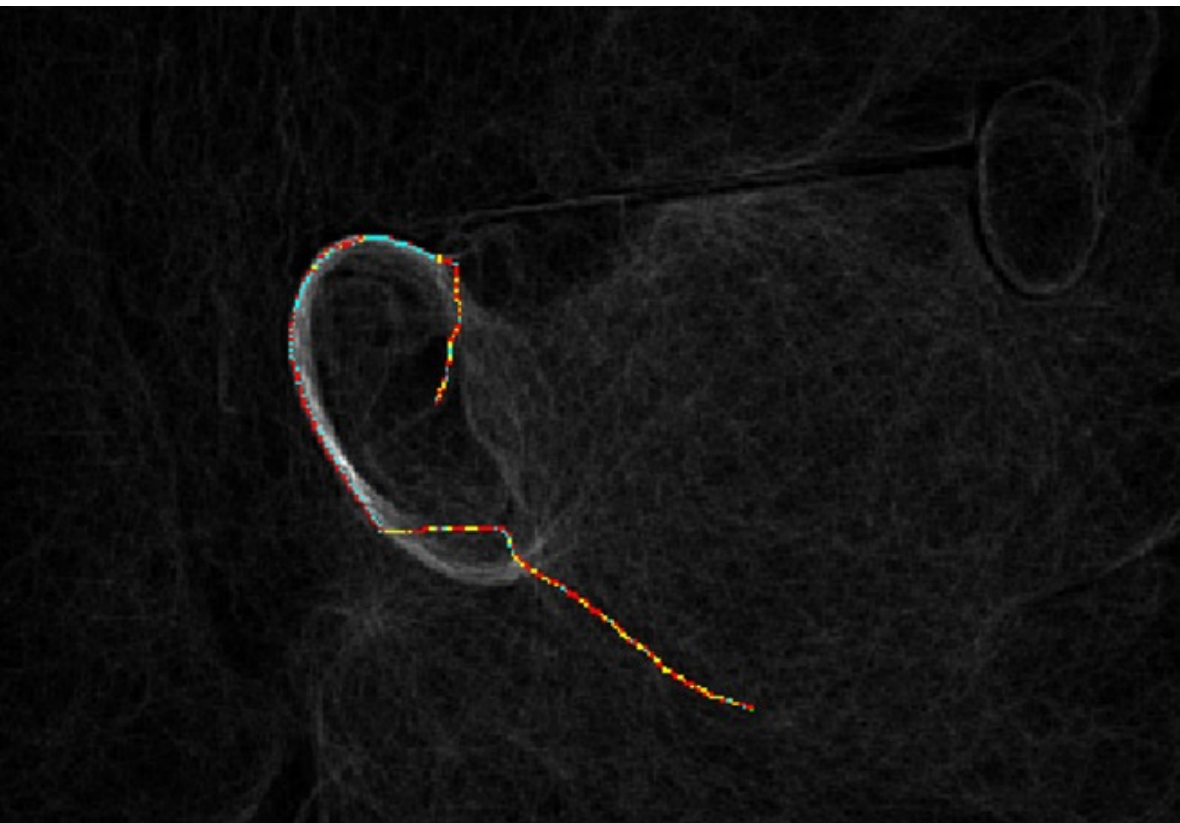


Medical Image Computing: Example Applications

- Application: Shape Analysis

DAVE MOSHER SCIENCE 11.12.10 04:32 PM

EARS COULD MAKE BETTER UNIQUE IDS THAN FINGERPRINTS



- **Application: Image Registration**
- Aligning images by warping
- How to define image similarity ?
 - Geometry-driven features?
Statistically ?
- How to define warp ?
 - Geometry ?
- How to optimize ?

